Varman: Research History & Empirical Evaluation of Adversarial Facial Privacy (2026)

A Comprehensive Retrospective on Project AMOR, Varman 2.0, and Varman 3.0

- **Date:** October 17, 2026
- **Author:** Varman Engineering & Research Pair Programming Team
- **Repository Branch:** `adaptive-strength`
- **Final Commit:** `c5a20a299d4920be69fe766cda1d493b7eea2494` [Measured]
- **Repository Path:** `<project_root>/Varman-backend`
- **Research Workspace:** `<project_root>/Varman-research`
- **Authoritative Handover:** `<user_home>/Downloads/HANDOVER.md`
- **Research Status: CLOSED as of October 17, 2026**

Executive Summary

This monograph documents the complete technical evolution, empirical findings, engineering failures, metric reconciliations, and final architecture of **Project Varman** (and its architectural predecessor, **AMOR 1.0**) across six months of active research from May 2026 to October 2026. The objective of the project was to develop a deployable facial privacy defense: an algorithmic system capable of perturbing a user's uploaded portrait photograph such that automated facial recognition scrapers fail to match it against enrollment databases, while preserving visual naturalness and surviving real-world social media compression (specifically JPEG compression across quality factors $Q=95$ down to $Q=50$).

Over three major architectural iterations, the project confronted fundamental machine learning realities:

1. **AMOR 1.0 (May 2026 – June 2026)** attempted full-canvas adversarial optimization (512×512 letterboxed images) wrapped in a complex microservices architecture (Celery, Redis, Caffe face detection). In white-box verification, AMOR was recorded in the handover notes (original test script not preserved, see GAP-01) as suffering a **100% failure rate** (0.00% evasion) [H]. Perturbations generated across the broad canvas were completely washed out by downstream Lanczos resizing, caused catastrophic visible banding, and exhibited intolerable inference latency (>15 s/image) on laptop GPUs.
2. **Varman 2.0 (June 2026 – October 2026)** pivoted to an ultralight, non-iterative CPU pipeline using YuNet face detection, facial landmark warping, and mid-frequency luminance Discrete Cosine Transform (DCT) noise. While computationally efficient (<150 ms execution) and aesthetically subtle (SSIM ≥ 0.90), empirical auditing against modern deep face recognition models recorded in the handover notes (original test script not preserved, see GAP-01) revealed an evasion rate of **approximately 0%** [H]. Modern deep embeddings are inherently invariant to subtle geometric warps and frequency shifts, rendering Varman 2.0 unprotective against automated scraping.
3. **Varman 3.0 / `adaptive-strength` (October 2026)** established the modern mathematical foundation of the defense: GPU-accelerated Expectation Over Transformation (EOT) Momentum-Iterative PGD (MI-FGSM, $\mu=0.90$) executed directly on canonical 112×112 aligned facial crops, incorporating differentiable frequency-domain JPEG simulation (`DiffJPEG`), luminance-only projection, and differentiable paste-aware inverse compositing back to native image coordinates.

Across extensive benchmarking on 150 LFW genuine pairs evaluated at a strict 1% False Acceptance Rate (FAR) across four distinct deep facial recognition models (FaceNet, ArcFace, SFace, MobileFaceNet), Varman 3.0 achieved substantial black-box protection:

- **Baseline Candidate F0** ($\varepsilon=8/255$, mask scale 1.00) achieved **36.39%** 4-model mean protection and **30.39%** held-out protection under real JPEG Q50 [Measured], rising to **42.03%** at JPEG Q90 [Measured]. In human exploratory screening across 10 source groups, F0 achieved a mean ordinal conspicuity rating of **1.30** on a 0–2 scale [Measured].
- **Production Preset F5** ($\varepsilon=8/255$, mask scale 1.15) achieved **45.73%** 4-model protection and **39.23%** held-out protection under real JPEG Q50 [Measured], rising to **47.84%** at JPEG Q90 [Measured]. In 10-group human screening, F5 achieved a mean rating of **1.50** [Measured], tying F0 in 8 out of 10 source groups while providing a +9.34 percentage point protection surge.
- **Maximum Protection Candidate F1** ($\varepsilon=8/255$, mask scale 1.30) achieved **49.95%** 4-model protection at Q50 and **55.59%** at Q90 [Measured], but produced severe boundary haloing, scoring **2.00** (Clear artifacts) across all 10 evaluated human screening groups [Measured].

The project established that under real JPEG compression at 1% FAR, the empirical black-box protection ceiling for canonical EOT-PGD is **roughly 50–56%** across four models [Measured]. Achieving higher evasion rates without severely compromising human aesthetic quality requires fundamentally different paradigms (such as generative identity hallucination or cross-image universal masks). With the integration of named preset `F5` and exhaustive multi-quality intermediate JPEG validation, Project Varman research is officially concluded and closed.

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Glossary of Scientific and Technical Terminology

- **False Acceptance Rate (FAR):** The statistical probability that an unconstrained biometric verification system erroneously classifies two facial photographs of different individuals as belonging to the same subject. In Varman evaluations, verification thresholds are calibrated on a dedicated calibration split (1,500 LFW impostor pairs) to enforce strict operating points of **1.0% FAR** and **0.1% FAR**.
- **Protection Rate / Flip Rate (%):** The primary defensive metric, defined as the proportion of genuine pairs (two distinct photos of the same subject) that successfully match when clean, but are driven below the calibrated 1% FAR threshold after protection is applied to one of the images. Flips are evaluated strictly over clean-baseline matches (valid denominator).
- **Surrogate vs. Held-Out Models:** In adversarial machine learning, a *surrogate model* is an architecture whose gradients are directly backpropagated to optimize the adversarial perturbation during training. A *held-out model* is a completely distinct architecture never exposed to the optimizer, whose weights and gradients remain strictly black-box. In Varman 3.0, **FaceNet** (InceptionResnetV1) and **ArcFace** (IResNet50) serve as dual differentiable surrogates, while **SFace** (OpenCV) and **MobileFaceNet** serve as non-differentiable held-out evaluation models. *(Note on Training Lineage: `w600k_r50` and `w600k_mbf` appear to share an upstream training lineage within the InsightFace release family [H]; 'held-out' denotes that the model was unseen during PGD and has a distinct architecture, not that it was trained on independent data.)*
- **Expectation Over Transformation (EOT):** A stochastic optimization technique wherein synthetic transformations (such as compression, spatial jitter, and photometric noise) are sampled dynamically during each step of gradient descent. In Varman 3.0, EOT samples random JPEG quality factors $Q \sim \mathcal{U}(50, 95)$ and spatial landmark displacements ± 2 px / $\pm 3%$ scale at each iteration.
- **Differentiable JPEG (`DiffJPEG`):** A GPU-accelerated PyTorch module that implements frequency-domain 8×8 block Discrete Cosine Transform, YCbCr color conversion, quantization table scaling, and Straight-Through Estimator (STE) rounding, allowing gradient backpropagation through lossy JPEG compression.
- **Momentum Iterative Fast Gradient Sign Method (MI-FGSM):** An iterative adversarial attack method that accumulates velocity vectors across optimization steps with decay factor $\mu=0.90$, stabilizing update directions and preventing oscillations in multi-surrogate loss landscapes.
- **Face SSIM Definition A (Inflated):** Structural Similarity Index computed across the full rectangular 112×112 canvas of the re-extracted composite crop. Because outer corners contain unperturbed clean pixels feathered into the background, Definition A artificially inflates SSIM (reporting ~0.9414 for baseline F0).
- **Face SSIM Definition B (Standardized Standard):** Structural Similarity Index evaluated strictly within the fixed canonical face mask (≥ 0.10, covering 8,095 pixels) on the aligned 112 crop, compared against the uncompressed source. Definition B isolates the true perceptual distortion on the subject's face independent of mask feathering geometry, and is standardized across all authoritative Varman reports (measuring 0.9015 for F0 and 0.8838 for F5).
- **Face SSIM Definition C (Pessimistic):** Structural Similarity Index evaluated directly on the raw rectangular PGD crop before feathered compositing back into the source photo (measuring 0.8753 for F0).
- **CIEDE2000 (ΔE_{00}):** The ISO/CIE standard color difference formula quantifying perceptual color shifts in CIELAB color space. $\Delta E_{00} \le 1.0$ is considered imperceptible to the human eye, while $\Delta E_{00} \ge 3.0$ indicates noticeable discoloration.
- **Newcombe-Wilson Paired Score Interval:** An asymptotic confidence interval for paired differences in binomial proportions ($p_\text{protected} - p_\text{control}$), providing robust statistical coverage over matched experimental units.
- **McNemar Exact Binomial Test:** A non-parametric paired statistical test evaluating whether discordant outcomes (n_{10} vs. n_{01}) between two conditions differ significantly from a fair coin toss ($p=0.5$).

Part I: Master Project Timeline

This timeline documents every major milestone, architectural pivot, empirical benchmark, bug discovery, and decision gate in the history of Project AMOR and Project Varman, spanning May 5, 2026 through October 17, 2026. Every entry is linked to an exact git commit hash or physical artifact on disk.

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1. Master Chronological Milestone Register

Date	Epoch / Version	Git Commit	Milestone / Event Description	Methodological Context & Technical Rationale	Quantitative Outcome / Evidence
2026-05-05	AMOR 1.0	`0686009`	Project AMOR Backend Initialization	Initial repository setup (`amor-backend`). Implemented microservice architecture using Celery, Redis broker, and Docker containerization.	Initial code commit; scaffolded API endpoints [M].
2026-05-07	AMOR 1.0	`bce4c1b`	2015 Caffe Face Detection Wiring	Implemented legacy OpenCV Caffe DNN face detector for test and exploration purposes.	Caffe prototxt and caffemodel integrated [M].
2026-05-09	AMOR 1.0	`8ff4a8e`	AMOR Redesign Phase v3	Restructured service layer into `services/` (DB, noise, recognition) and `utils/` .	Architectural refactoring [M].
2026-05-12	AMOR 1.0	`17d699b`	Full-Canvas PGD Optimization Pipeline	Wired full-canvas PGD optimizer on 512×512 letterboxed images against FaceNet and ArcFace surrogates.	Endpoints exposed in FastAPI `main.py` [M].
2026-05-24	AMOR 1.0	`9500f22`	White-Box Testing Phase 1 Execution	Automated test suite execution across image protection routes.	**59 test failures** logged in test report [M].
2026-05-25	AMOR 1.0	`6e47ae7`	White-Box Verification Failure Confirmed	Evaluation of protected images against face recognition models. Perturbation failed to survive downsampling.	**100% failure rate** recorded in commit log [M].
2026-06-13	Varman 1.0	`c78297f`	Architecture Pivot: AMOR $\to$ Varman	Decision to abandon heavy full-canvas optimization; repository transitioned to Varman MVP architecture.	Formal retirement of AMOR codebase [M].
2026-06-16	Varman 1.0	`8d78f46`	Removal of Heavy Processing Services	Stripped out Celery, Redis, and heavy PyTorch dependencies to reduce latency.	Lightweight backend restructuring [M].
2026-06-30	Varman 1.0	`7c964b2`	Varman 1.0 Release Candidate	Finalization of Varman Phase 1 backend architecture with JWT auth and SQLite database.	Phase 1 backend completed (`fc9a0d3`) [M].
2026-08-08	Varman 2.0	`7ec6d4b`	Pivot to Lightweight ONNX/OpenCV Engine	Shift to real-time, non-iterative CPU pipeline using facial landmark warping and DCT mid-frequency perturbation.	Replaced GPU PGD with CPU filters [M].
2026-08-09	Varman 2.0	`ad1709e`	YuNet ONNX Face Detector Integration	Integrated `face_detection_yunet_2023mar.onnx` into `models/` for 5-point landmark detection.	YuNet ONNX integrated (232.5 KB) [M].
2026-08-09	Varman 2.0	`4ebadab`	Phase 2 Validation Harness Added	Added benchmark harness bridging API routes with synthetic face inputs and SSIM evaluation.	Verified Global SSIM ≥ 0.90 [M].
2026-08-11	Varman 2.0	`0a6f445`	Preset Re-architecture	Overhauled protection presets: Light (warp 0.005, DCT 3.0), Balanced (warp 0.015, DCT 10.0), Aggressive (warp 0.025, DCT 18.0).	Guardrail target SSIM 0.85–0.95 [M].
2026-09-16	Varman 2.0	`2203ab4`	Production Database Support Added	Added async MySQL connection pooling alongside SQLite storage.	Pylance type safety and async drivers [M].
2026-10-09	Varman 2.0	`11f7894`	The Zero-Protection Discovery	Rigorous black-box evaluation against ArcFace, FaceNet, SFace, and MobileFaceNet revealed v2.0 had no protective effect.	**~0% protection** against deep models [M].
2026-10-09	Varman 3.0	`11f7894`	Evaluation Suite & Testbed Overhaul	Refactored `evaluate_face_models.py` with dynamic CUDA ONNX provider, Wilson score CIs, Newcombe paired intervals, and calibrated thresholds.	Thresholds calibrated on 1,500 impostors [M].
2026-10-10	Varman 3.0	`63d8aa9`	Benchmarking Tooling Hardened	Added live ETA logging and graceful signal handling; established `adaptive-strength` branch.	Pre-sweep checkpoint created (`5eba251`) [M].
2026-10-11	Varman 3.0	`d4e52be`	Real-Photo & LFW Baseline Sweep	Evaluated luminance-only PGD at $\varepsilon \in {6, 7, 8}/255$. Epsilon 8 achieved 21.6% LFW flip rate and 25.0% on real photos.	Baseline flip rate 21.6% (1% FAR PNG) [M].
2026-10-12	Varman 3.0	`490e8b3`	PGD Step Convergence Benchmark	Tested 25, 50, and 75 steps under fixed and cosine step schedules. 50/75 steps gained only +1–2 pp and degraded Q50 survivability.	25 steps selected as Pareto optimal [M].

Date	Epoch / Version	Git Commit	Milestone / Event Description	Methodological Context & Technical Rationale	Quantitative Outcome / Evidence
2026-10-13	Varman 3.0	`ab57b83`	Calibration Evidence Preserved	Copied threshold calibration evidence (`calibrated_thresholds_sweep.json`) into versioned directory.	Thresholds formalized at 1% / 0.1% FAR [M].
2026-10-13	Varman 3.0	`feeb23d`	Hinge Loss Experimentation	Tested threshold-aimed hinge loss with 15% margin. Protection dropped from 22.70% to 17.93% (4-model mean).	Hinge loss rejected as ineffective [M].
2026-10-13	Varman 3.0	`b20de80`	Surrogate Weight Ratio Sweep	Tested $w_\text{fn} \in {0.10, 0.25, 0.50}$. Decreasing FaceNet weight caused severe protection collapse (14.62% at 0.10).	Low FaceNet weights rejected [M].
2026-10-13	Varman 3.0	`22c02c0`	Surrogate Weight Follow-up ($w=1.0$)	Evaluated $w_\text{fn} \in {0.50, 1.0, 2.0}$. $w=1.0$ produced highest 4-model mean (25.50% PNG, 21.28% Q50).	$w=1.0$ established as primary setting [M].
2026-10-13	Varman 3.0	`22c02c0`	40-Pair Pipeline Stage Diagnostic	Tracked perturbation decay across pipeline stages. Discovered that 50–60% of adversarial disruption is lost during inverse compositing.	Optimizer inside: 100% ArcFace / 87.5% FaceNet $\to$ Post-paste: 47.5% / 42.5% [M].
2026-10-14	Varman 3.0	`5e89d2d`	External Chameleon Research Audit	Analyzed Chameleon codebase (`D:\JavaProjects\Varman-research\external\Chameleon`). Contrasted universal masks with per-image EOT.	Audit reports generated in reports folder [M].
2026-10-14	Varman 3.0	`7268342`	Paste-Aware Optimization Breakthrough	Differentiable paste-aware PGD (`run_paste_aware_experiment.py`). Row 2 gained +9.12 pp at Q50 (30.22%); Row 3 (with jitter) reached 36.57%.	Q50 protection surge: 21.10% $\to$ 36.57% [M].
2026-10-15	Varman 3.0	`14e61f0`	YuNet Landmark Drift Controlled Pilot	Measured real landmark displacement on 30 pairs. Mean drift 1.33–1.49 px (p95 ~2.8 px). Refuted landmark drift collapse hypothesis.	Landmark drift confirmed negligible [M].
2026-10-15	Varman 3.0	`92ac31e`	Region-Size Ablation Experiment	Evaluated mask scales 0.70, 1.00, and 1.30 across 150 pairs. Scale 1.30 at $\varepsilon=8$ pushed Q50 protection to 48.36% (4-model) / 40.64% (held-out).	Discovered Face SSIM metric discrepancy between scripts [M].
2026-10-15	Varman 3.0	`7eaf0e5`	Metric Reconciliation & Def B Standard	Traced discrepancy to full-canvas feathering inflation. Formally standardized Definition B (fixed canonical mask ≥ 0.10 on aligned 112 crop vs uncompressed source).	Standardized Definition B established [M].
2026-10-15	Varman 3.0	`7eaf0e5`	Coverage Frontier Benchmark (F0–F5)	Executed authoritative 6-condition frontier on 150 pairs: F0 (36.39%), F1 (49.95%), F2 (37.10%), F5 (45.73%). Evaluated graded grain on 50 pairs.	Dose-response confirmed; F2 matches F0 protection with lower CIEDE p95 [M].
2026-10-16	Varman 3.0	`7eaf0e5`	Full `human_test_v3` Package Built	Assembled 225-stimulus, 45-source group package (`ST_000001` – `ST_000225`) across 5 conditions with blind web viewer.	19 automated validation checks passed [M].
2026-10-16	Varman 3.0	`7eaf0e5`	50-Stimulus SSIM Quick Review Built	Sampled 10 source groups (`ST_000226` – `ST_000275`) sorted by descending SSIM to reduce manual evaluation load.	12 automated checks passed; SHA verified [M].
2026-10-16	Varman 3.0	`7eaf0e5`	50-Stimulus Quick Review Analyzed	Evaluator completed 50 ratings: Unprotected 0.00, F0 1.30, F2 1.70, F1 2.00, F1 Graded 2.00. F0 provisionally preferred over F2.	F0 achieved lowest conspicuousness among protected options [M].
2026-10-16	Varman 3.0	`7eaf0e5`	F0 vs. F2 Arithmetic Reconciliation	Audited discrepancy in paired counts. Reconciled from raw CSV: F0 cleaner in 5 groups, F2 in 1, tied in 4.	Arithmetic error corrected in reports [M].
2026-10-16	Varman 3.0	`7eaf0e5`	F5 Add-on Visual Screening	Evaluated F5 on same 10 source groups: Mean rating 1.50 (5 Slight, 5 Clear). Tied F0 in 8 groups, worse in 2, but +9.34 pp more protective.	Disproved Face SSIM correlation with human ratings [M].
2026-10-16	Varman 3.0	`7eaf0e5`	Intermediate JPEG Validation (Q95–Q50)	Re-encoded pre-JPEG outputs across Q95, 90, 75, 70, 60, 50. Confirmed protection monotonicity (peaks at Q90–95, drops smoothly to Q50).	Protection curves validated across quality spectrum [M].
2026-10-16	Varman 3.0	`c5a20a2`	F5 Named Preset Integrated	Re-added F5 preset into `StrongProtectionEngine` and `ProtectionEngine` . Added config unit test (`test_f5_preset.py`).	Test suite passed 100% in 4.00s [M].
2026-10-17	Varman 3.0	`c5a20a2`	Research Closure	Created comprehensive research history monograph, updated README, and closed research.	Research officially closed [M].

2. Key Decision Register

```
flowchart TD
    D1["Decision 1: Dual Execution Modes<br>(Fast CPU vs. Strong GPU)"] --> D2["Decision 2: 1% FAR Biometric Benchmark Standard"]
    D2 --> D3["Decision 3: Real JPEG Compression Verification<br>(Never DiffJPEG proxy alone)"]
    D3 --> D4["Decision 4: Luminance-Only Perturbation<br>(chroma_scale = 0.0)"]
    D4 --> D5["Decision 5: Rejection of Hinge Loss & Low Surrogate Weights"]
    D5 --> D6["Decision 6: Differentiable Paste-Aware Optimization"]
    D6 --> D7["Decision 7: Definition B Face SSIM Standardization"]
    D7 --> D8["Decision 8: Selection of F5 as Primary Production Preset"]
```

1. **Dual Execution Modes (No Silent Auto-Routing):** The engine exposes two explicit modes: `mode="fast"` (non-protective legacy CPU engine) and `mode="strong"` (GPU EOT-PGD). If CUDA is unavailable when strong mode is requested, the system falls back to fast mode but explicitly sets metadata flags (`requested_mode="strong"` , `fallback_to_fast=True` , `fallback_reason="..."`). Auto-routing without user visibility was rejected to preserve transparency.
2. **Strict 1% FAR Evaluation Standard:** All face recognition benchmarks must evaluate protection rates at a fixed operating threshold corresponding to **1.0% False Acceptance Rate** (with secondary checks at 0.1% FAR) on an independent impostor calibration split. Reporting disruption rates without declaring the operating threshold or claiming protection against "all scrapers" was banned.
3. **Physical File Verification Gate:** Synthetic gradients from differentiable JPEG approximations (`DiffJPEG`) are permitted only inside the optimization loop. All empirical benchmark evaluations must save real physical JPEG files to disk, reload them, detect faces independently, and score embeddings on reloaded bytes.
4. **Luminance-Only Optimization:** Color perturbation was constrained strictly to the Y (luminance) channel in YCbCr space ($\text{chroma_scale} = 0.0$). This reduced perceptual color shift (ΔE_{00}) by ~40% relative to full-RGB perturbation at identical epsilon budgets without sacrificing protection.
5. **Rejection of Hinge Loss:** The threshold-aimed hinge loss was formally rejected after empirical testing proved it caused premature gradient starvation on ArcFace, reducing 4-model protection by 4.77 percentage points.
6. **Differentiable Inverse Compositing:** The optimization forward pass must model the full inverse affine warp, feathered elliptical mask blending, and re-alignment inside the backpropagation loop (`run_paste_aware_experiment.py`), eliminating the 50% post-paste degradation rebound.
7. **Definition B Face SSIM Standardization:** Objective facial structural similarity must be computed strictly within the fixed canonical face mask (≥ 0.10) on the aligned 112 crop, compared against the uncompressed source, eliminating the ~0.04 to ~0.06 SSIM artificial inflation caused by unperturbed corner pixels in Definition A.
8. **Selection of Preset F5:** Candidate F5 ($\varepsilon=8/255$, mask scale 1.15) was selected as the final named production preset. It bridges the gap between F0 and F1, delivering a +9.34 percentage point protection improvement over F0 under real JPEG Q50 (45.73% vs. 36.39% [Measured]) while avoiding the severe, full-face boundary haloing of F1 (rating 1.50 vs. 2.00 [Measured]).

Part II: AMOR 1.0 — The Monolithic Full-Canvas Prototype (May 2026 – June 2026)

1. Original Objectives and Threat Model

Project **AMOR** (occasionally referenced in legacy commits as *Armor*) was initiated on May 5, 2026 (commit `0686009` : `"Project AMOR : backend"` [Measured]). The core objective was to create an automated, asynchronous facial obfuscation platform capable of protecting user portrait photographs against unauthorized biometric harvesting and unconstrained web scrapers.

The conceptual threat model envisioned by AMOR 1.0 was broad:

- **Attacker Profile:** Mass web-scraping pipelines and commercial facial recognition databases crawling social media platforms to extract biometric identity representations without consent.
- **Defensive Mechanism:** Additive adversarial perturbation generated via Projected Gradient Descent (PGD) to push facial embeddings beyond matching thresholds while maintaining acceptable perceptual image quality.
- **Operating Environment:** A multi-container Dockerized microservices backend accessible via REST API endpoints, with background task offloading for computationally intensive gradient calculations.

---

2. System Architecture & Engineering Stack

The AMOR 1.0 architecture was designed around a distributed microservice pattern typical of enterprise web services. However, this architectural complexity was introduced before the underlying adversarial machine learning mechanics had been empirically validated.

```
flowchart TD
    Client["Client / Web UI"] -->|HTTP POST /protect| API["FastAPI Gateway (main.py)"]
    API -->|Enqueue Task| Redis["Redis Message Broker"]
    Redis -->|Dispatch Job| Celery["Celery Distributed Worker (celery_worker.py)"]
    Celery -->|Face Detection| Caffe["Caffe DNN Face Detector (2015 SSD)"]
    Caffe -->|Letterbox 512x512| PGD["Full-Canvas PyTorch PGD Engine"]
    PGD -->|Gradient Backprop| Surrogates["Surrogates (FaceNet + ArcFace)"]
    PGD -->|Lanczos Upscale| Composite["Reconstructed Image"]
    Composite -->|Write Output| Storage["Temporary Storage / DB"]
```

2.1 The Infrastructure Stack

As documented in commits `17d699b`, `b35f9c9`, and `a77939f` [Measured], the AMOR 1.0 repository structure was organized into distinct service tiers:

- `main.py`: FastAPI application gateway defining HTTP endpoints for user registration, image upload, protection status polling, and download.
- `celery_worker.py`: Distributed task execution worker using Celery to pull asynchronous protection jobs from a message queue.
- `services/`:
  - `services/recognition/`: Face detection and embedding extraction wrapper.
  - `services/noise/`: Adversarial perturbation generation logic.
  - `services/db/`: Database models and persistence layer.
- `storage/`: Temporary on-disk staging directory for raw and processed image uploads.
- `config.py`: Environment variable configuration and secret management.
- `Dockerfile` & `docker-compose.yml`: Container orchestration definitions linking FastAPI, Celery, and a Redis message broker.

2.2 Face Detection Pipeline

In commit `bce4c1b` (May 7, 2026: `"implemented 2015 Caffe model for face detection"` [Measured]), the system incorporated a legacy single-shot multibox detector (SSD) implemented in Caffe (`res10_300x300_ssd_iter_140000.caffemodel` with prototxt configuration).

- **Execution:** Images were decoded via OpenCV and passed to `cv2.dnn.readNetFromCaffe`.
- **Limitation:** The 2015 Caffe detector extracted only crude rectangular bounding boxes $[x, y, w, h]$ with a low-confidence threshold (~0.5). Crucially, it provided **no facial landmark coordinates** (eyes, nose, mouth corners), making canonical similarity alignment mathematically impossible.

2.3 The Full-Canvas 512x512 PGD Optimizer

Lacking facial landmark alignment, AMOR 1.0 attempted to solve adversarial perturbation at the global image level:

1. **Letterboxing:** Input photographs of arbitrary aspect ratio and resolution (e.g., 1000×1500 px) were padded and resized down to a uniform square canvas of 512×512 pixels using bilinear or bicubic downsampling.
2. **Surrogate Objectives:** The full 512×512 image was fed simultaneously into two deep recognition models:
   - A PyTorch implementation of **FaceNet** (InceptionResnetV1).
   - An early **ArcFace** ResNet backbone.
3. **Loss Function:** An unweighted cosine similarity minimization objective: $$\mathcal{L}*text{AMOR}* = *cos(f*\text{AF}(x_\text{adv}), f_\text{AF}(x_\text{clean})) + \cos(f_\text{FN}(x_\text{adv}), f_\text{FN}(x_\text{clean}))$$
4. **Optimization Loop:** Standard L_∞-bounded PGD run across the entire $512 \times 512 \times 3$ tensor for 10–20 iterations at step size $\alpha \approx \varepsilon / 10$, using an epsilon budget of $\varepsilon \in [8/255, 16/255]$.
5. **Reconstruction via Lanczos Upsampling:** Following optimization, the perturbed 512×512 tensor was converted back to an 8-bit image and upsampled back to the original photograph dimensions using Lanczos interpolation.

---

3. Empirical Verification and The 100% Failure Rate

In late May 2026, the author implemented an automated white-box test harness to measure whether the generated images successfully evaded facial recognition. As recorded in the handover notes (original test script not preserved, see GAP-01), the results revealed complete evasion failure:

- **May 24, 2026 (Commit `9500f22`):** `"AMOR white box testing phase 1 results 59 fails remark : testing in progress"` [Measured].
- **May 25, 2026 (Commit `6e47ae7`):** `"white box testing failure rate is 100%"` [Measured].

Every protected photograph produced by the AMOR 1.0 pipeline was recorded as matched by the target face recognition models when re-evaluated at standard verification thresholds, achieving **zero protection (0.00% evasion)** [H].

---

4. Root Cause Failure Analysis

Why did AMOR 1.0 fail completely? Detailed retrospective analysis reveals five severe architectural flaws:

4.1 The Resampling Low-Pass Filter (The Washing Effect)

The most fatal flaw in AMOR 1.0 was optimizing on a resized 512×512 canvas and then resizing back to native resolution:

- Adversarial perturbations generated by gradient descent against convolutional neural networks are predominantly composed of high-spatial-frequency noise patterns.
- Downsampling an original photo (e.g., 1200×1600) to 512×512 discarded fine facial texture.
- More destructively, upsampling the perturbed 512×512 output back to native dimensions via Lanczos interpolation acted as a powerful low-pass smoothing filter. The high-frequency adversarial gradients were completely smoothed away, restoring the underlying clean facial geometry and allowing face recognition models to match the identity with ease.

4.2 Spatial Gradient Dispersion Across Non-Face Pixels

Because PGD optimized the entire 512×512 canvas rather than isolating the face:

- Gradients flowed into clothing, hair, and background regions.
- The optimizer frequently satisfied loss minimization by generating massive, conspicuous artifacts in plain background regions while leaving the core discriminative facial features (eyes, nose, lip contours) largely unperturbed.
- This resulted in an unacceptably low global Structural Similarity Index (SSIM < 0.80), causing visible grain and blotchy chromatic banding across the background while providing zero facial defense.

4.3 Absence of Differentiable Compression (The JPEG Guillotine)

AMOR 1.0 did not model image compression during optimization. The forward pass computed gradients directly on uncompressed floating-point tensors:

- When the resulting image was saved to disk as a standard JPEG file or re-encoded by social media platforms, standard 8×8 block DCT quantization immediately eliminated the subtle residual pixel perturbations.
- The perturbation lived only in unquantized 32-bit floating-point RAM and was destroyed upon physical disk serialization.

4.4 Uncalibrated Verification Thresholds

Early AMOR documentation claimed that "disruption was achieved" whenever cosine similarity dropped below arbitrary thresholds (such as 0.44 for FaceNet or 0.41 for ArcFace) [Measured].

- In reality, as later proven in Project Varman, 0.4427 was merely FaceNet's 1% FAR threshold on LFW genuine pairs, not an evasion score.
- Because clean genuine faces often match with similarities between 0.70 and 0.85, dropping similarity to 0.50 does not prevent a match if the verification threshold is set to 0.3754. AMOR 1.0 lacked an empirical impostor calibration split, resulting in completely invalid claims of success.

4.5 Extreme Latency and Infrastructure Overhead

On the developer's laptop hardware (NVIDIA GeForce RTX 2050 Mobile with 4 GB VRAM), executing 20 PGD iterations over a full $512 \times 512 \times 3$ tensor alongside two deep backbones took **15 to 25 seconds per image** [Measured].

- Peak VRAM frequently exceeded 3.5 GB, causing CUDA out-of-memory errors when processing larger images.
- Running Celery, Redis, and multi-container Docker daemons consumed valuable system RAM and disk storage, creating immense developer friction during testing.

---

5. Deprecation and Architecture Pivot

By early June 2026, the developer recognized that the monolithic full-canvas approach was fundamentally flawed:

1. Full-canvas optimization could never survive multi-scale resizing.
2. Background PGD was visually conspicuous and computationally wasteful.
3. Heavy microservice infrastructure obscured underlying algorithmic failures.

On **June 13, 2026**, the developer executed commit `c78297f`:
`"chore: pivot from AMOR to Varman MVP architecture"` [Measured].

This commit marked the formal termination of Project AMOR and the birth of **Project Varman**. Over the subsequent three days, the Celery workers, Redis queues, and Caffe detectors were excised from the repository (`8d78f46`: `"Refactor: Remove image processing and noise services"` [Measured]), clearing the way for a lightweight, standalone architecture focused directly on the mathematics of facial protection.

Part III: Varman 2.0 — The Non-Iterative CPU Heuristic Engine (June 2026 – October 2026)

1. Design Philosophy: The Quest for Real-Time CPU Privacy

Following the collapse of AMOR 1.0 in June 2026, Project Varman was conceived as a radical paradigm shift. The core hypothesis driving Varman 2.0 was that iterative adversarial gradient descent (PGD) was inherently too slow, too fragile, and too computationally demanding for practical deployment on consumer hardware or edge servers.

The design philosophy of Varman 2.0 was guided by three foundational tenets:

1. **Zero GPU Requirement:** The engine must execute entirely on standard multi-core CPUs using lightweight OpenCV and ONNX Runtime CPU execution providers. Heavy deep learning frameworks (PyTorch, TensorFlow) were completely removed from production runtime (`583f445` : `"Cleanup: remove legacy PyTorch references and unneeded benchmarks"` [Measured]).
2. **Sub-200ms Latency:** Processing must be instantaneous (<200 ms per photograph), enabling real-time client uploads without asynchronous queuing.
3. **Guaranteed Aesthetic Fidelity:** Structural similarity must be strictly guaranteed via automated quality guardrails, ensuring that Global SSIM never drops below 0.90.

This philosophy culminated in the implementation of the **Varman Dedicated Lightweight Protection Engine** in August 2026 (commits `7ec6d4b` , `ad1709e` , `4ebadab` , and `0a6f445` [Measured]).

---

2. The Varman 2.0 Architectural Pipeline

The Varman 2.0 pipeline replaced iterative optimization with a deterministic, five-stage analytical image processing pipeline implemented in `app/protection/protection_engine.py`:

```
flowchart TD
    Input["Input Image (BGR)"] --> S1["Stage 1: YuNet Face & Landmark Detection (ONNX)"]
    S1 -->|5-Point Landmarks| S2["Stage 2: Relative Soft-Masked Landmark Face Warping"]
    S2 -->|Warped Face| S3["Stage 3: Y-Channel Mid-Frequency DCT Perturbation"]
    S3 -->|Optional Layer| S4["Stage 4: Low-Intensity Spatial Grid Layer"]
    S4 -->|Candidate Output| S5["Stage 5: Adaptive SSIM Quality Guardrail (Bisection)"]
    S5 -->|Passes SSIM Floor| Output["Protected Image Output (PNG)"]
    S5 -->|Fails SSIM Floor| Attenuate["Attenuate Warp & DCT by 0.85x"] --> S5
```

2.1 Stage 1: YuNet Face and 5-Point Landmark Detection

To eliminate the crude bounding boxes of the 2015 Caffe detector, Varman 2.0 integrated the state-of-the-art **YuNet** face detector (`face_detection_yunet_2023mar.onnx` , 232,589 bytes, SHA-256 `8f2383e4dd...` [Measured]):

- **Execution:** Loaded via OpenCV's native DNN module: `cv2.FaceDetectorYN.create()` .
- **Inference Size:** Input images were dynamically scaled to $(300, 300)$ or native resolution, with a score threshold of 0.60 and non-maximum suppression (NMS) threshold of 0.30.
- **Outputs:** Extracted a 15-float detection vector per face containing:
  - Bounding box coordinates: $[x, y, w, h]$
  - 5-point facial landmark coordinates: Right Eye (x_0, y_0), Left Eye (x_1, y_1), Nose Tip (x_2, y_2), Right Mouth Corner (x_3, y_3), Left Mouth Corner (x_4, y_4)
  - Detection confidence score

If no face was detected, the engine gracefully skipped facial warping and proceeded directly to whole-image luminance DCT filtering, ensuring the system never crashed on non-face uploads.

2.2 Stage 2: Relative Soft-Masked Landmark Face Warping

The core geometric distortion in Varman 2.0 was designed to alter facial proportions relative to the subject's natural geometry:

1. **Displacement Vector Calculation:** For each of the five detected landmarks, an outward radial displacement vector was computed proportional to the face bounding box width: $$\Delta \vec{p}*i* = *\text{warp_scale}* *\cdot* *w*\text{face} \cdot \vec{d}_i$$ where $\vec{d}_i$ represents pre-configured directional unit vectors designed to subtly stretch eye-to-eye and nose-to-mouth distances.
2. **Dense Motion Field:** A smooth 2D displacement field (D_x, D_y) was generated across the facial bounding box by applying an inverse-distance weighting (IDW) or Gaussian kernel centered at the five landmarks.
3. **Soft Elliptical Masking:** The displacement field was modulated by an elliptical Gaussian falloff mask centered on the face: $$M(x, y) = \exp\left(-\left(\frac{(x - c_x)^2}{2\sigma_x^2} + \frac{(y - c_y)^2}{2\sigma_y^2}\right)\right)$$ This ensured that warping decayed smoothly to zero at the jawline, cheeks, and forehead, preventing discontinuous tearing or jagged seams with the surrounding unperturbed canvas.
4. **Remapping:** The image was resampled using bilinear backward mapping via `cv2.remap()`.

2.3 Stage 3: Y-Channel (Luminance) Mid-Frequency DCT Perturbation

To complement geometric warping with frequency-domain disruption without causing conspicuous chromatic shifts, Varman 2.0 implemented block Discrete Cosine Transform filtering:

1. **Color Conversion:** The image was converted from BGR to YCrCb space, isolating the luminance channel (Y).
2. **8×8 Block Partitioning:** The luminance channel was partitioned into non-overlapping 8×8 pixel blocks.
3. **Forward 2D DCT:** Each block was transformed into frequency space using `cv2.dct()`: $$F(u, v) = \text{DCT}(f(x, y)), \quad u, v \in [0, 7]$$
4. **Mid-Frequency Energy Injection:** Perturbation was injected exclusively into the mid-frequency band ($u + v \in [3, 7]$), leaving the DC component ($u=v=0$) and low frequencies untouched to preserve overall lighting, while avoiding high frequencies ($u + v > 7$) that are heavily quantized by JPEG encoders. $$F'(u, v) = F(u, v) + \text{dct_strength} \cdot S(u, v) \cdot \operatorname{sgn}(F(u, v))$$ where $S(u, v)$ is a mid-frequency bandpass mask.
5. **Inverse DCT & Clipping:** Each block was reconstructed via `cv2.idct()`, clamped to $[0, 255]$, and reassembled into the full luminance plane.

2.4 Stage 4: Optional Spatial Grid Layer

As an optional defensive layer, Varman 2.0 included a low-intensity spatial grid overlay (`enable_grid=True`). The grid injected a subtle, periodic sinusoidal lattice pattern across the luminance channel: $$G(x, y) = A_\text{grid} \cdot \sin\left(\frac{2\pi x}{T}\right) \cdot \sin\left(\frac{2\pi y}{T}\right)$$ with spatial period $T = 16$ px and amplitude $A_\text{grid} \in [1.0, 2.5]$ intensity units. The hypothesis was that periodic spatial frequencies would disrupt the spatial pooling operations of convolutional neural networks.

2.5 Stage 5: Adaptive SSIM Quality Guardrail

To ensure the engine never output visibly degraded photographs, Varman 2.0 introduced an iterative attenuation guardrail:

1. **Target Invariants:** Each preset specified two quality floors:
   - Target Global SSIM (≥ 0.85 to 0.95) evaluated across the entire image.
   - Target Face SSIM (≥ 0.85 to 0.95) evaluated across an expanded $1.2\times$ bounding box centered on the primary face.
2. **Fast SSIM Computation:** Structural similarity was computed rapidly using Gaussian-weighted windowing on luminance channels via an optimized OpenCV implementation.
3. **Attenuation Loop:** If the candidate protected image failed either SSIM floor, an attenuation factor λ (initially 1.0) was decayed: $$\lambda \leftarrow \lambda \cdot 0.85$$ The applied warp scale and DCT strength were scaled by λ: $$\text{warp_scale}*text{applied}* = *lambda* *cdot* *text{warp_scale}, *quad* *text{dct_strength}*\text{applied} = \lambda \cdot \text{dct_strength}$$ and the image was re-synthesized.
4. **Termination:** The bisection search repeated until both SSIM constraints were satisfied, or until $\lambda \le 0.10$ (at which point the engine set `guardrail_satisfied=False` and reverted to the lowest attenuation state).

---

3. Preset Definitions and Configurations

Varman 2.0 formalized three named operating presets in `app/protection/protection_engine.py` (lines 26–45 [Measured]):

Preset Name	Warp Scale	DCT Strength	Target Global SSIM	Target Face SSIM	Primary Design Intent
Light	0.005 (0.5% face width)	3.0	0.95	0.95	Quality-first mode; visually imperceptible adjustments.
Balanced	0.015 (1.5% face width)	10.0	0.90	0.90	Default operating point; balanced geometric and frequency shift.
Aggressive	0.025 (2.5% face width)	18.0	0.85	0.85	Maximum allowable disruption within aesthetic acceptability.

4. Production Integration & Verification Harness

Between August and September 2026, Varman 2.0 was fully productionized into a complete enterprise application:

- **FastAPI Endpoints:** `POST /api/v1/images/protect` wired in `app/routes/images.py`, supporting multipart file uploads, preset selection, optional grid toggles, and metadata response serialization.
- **Async Database Architecture:** Added MySQL connection pooling via `asyncmy` / `aiomysql` and SQLite fallback in `app/database.py` (commit `2203ab4` [Measured]). Database migrations tracked user accounts, upload timestamps, original vs. protected image sizes, and processing latency.
- **Phase 2 Validation Harness:** Implemented in `benchmarks/phase2_validation.py` and tested via `tests/test_engine.py` and `tests/test_phase2_validation.py` [Measured].

4.1 Measured Operational Performance

On the developer's laptop CPU (Intel Core i7-12700H, Windows 11), Varman 2.0 achieved extraordinary computational performance [Measured]:

- **Inference Latency: 18.5 ms** per 256×256 face image; **112.4 ms** per 1080×1920 full HD photograph.
- **Memory Footprint:** Peak RAM consumption under 85 MB; zero GPU VRAM allocation.
- **Guardrail Convergence:** $100%$ of synthetic and natural test images satisfied the Balanced preset guardrail ($\text{SSIM} \ge 0.90$) within 0 to 2 attenuation steps.
- **Subjective Naturalness:** Informal visual inspection by the project author and colleagues confirmed that protected images appeared completely natural, with no visible pixelation, chromatic noise, or obvious signs of synthetic tampering.

---

5. The Baseline-Corrected Y-NPR Metric and Its Flaws

To quantify frequency-domain perturbation survival, the Varman 2.0 team devised a custom metric titled **Baseline-Corrected Y-Channel Noise Preservation Ratio (Y-NPR)**.

- **Definition:** The ratio of high-frequency energy in the protected image surviving JPEG compression ($Q=75$) relative to the energy loss in the clean original image: $$\text{Y-NPR} = \frac{\Delta E_\text{prot}(Q75) - \Delta E_\text{clean}(Q75)}{\Delta E_\text{prot}(\text{PNG}) + \epsilon} \times 100\%$$
- **Reported Claims:** Early validation reports cited Y-NPR scores between 72% and 88%, claiming this mathematically proved that Varman's mid-frequency DCT noise survived compression.
- **Methodological Flaw & Retirement:** Subsequent mathematical auditing in October 2026 revealed that Y-NPR was fundamentally flawed:
  1. In the codebase, Y-NPR values were hard-clamped at 100.0%, hiding overflow anomalies caused by compression edge artifacts.
  2. High-frequency energy preservation does not correlate with adversarial feature disruption in neural networks. Noise can survive JPEG compression completely while being completely ignored by face recognition models.
  3. Consequently, Y-NPR was formally **retired** and excluded from all authoritative scientific reporting (Handover Section 5, line 104 [Measured]).

---

6. The Zero-Protection Discovery (October 9, 2026)

On October 9, 2026, the author implemented a rigorous, black-box face recognition evaluation suite (`benchmarks/evaluate_face_models.py` , commit `11f7894` [Measured]). Rather than relying on SSIM or custom noise metrics, the suite evaluated whether protected images actually prevented deep face recognition models from matching genuine identities.

The evaluation suite tested four state-of-the-art deep facial recognition backbones:

1. **FaceNet** (InceptionResnetV1, trained on VGGFace2)
2. **ArcFace** (IResNet50 / ResNet50, InsightFace family)
3. **SFace** (OpenCV FaceRecognizerSF, trained on broader Asian/Caucasian sets)
4. **MobileFaceNet** (Edge lightweight embedding network)

Thresholds were calibrated to enforce a strict **1.0% False Acceptance Rate (FAR)** using 1,500 LFW impostor pairs.

6.1 The Zero-Protection Outcome

As recorded in the handover notes (original test script not preserved, see GAP-01), when 150 LFW genuine pairs were protected using Varman 2.0 (Balanced and Aggressive presets) and evaluated against the four models, the flip rate was: $$\text{Protection Rate (1% FAR)} \approx \mathbf{0.0%} \quad [H]$$

- Across all 150 pairs, **zero models were disrupted** [H].
- Cosine similarity between clean enrollment photos and protected photos remained between 0.68 and 0.82, sitting overwhelmingly above the 1% FAR thresholds (ArcFace 0.1476, FaceNet 0.3754, SFace 0.2920, MobileFaceNet 0.1688) [H].
- Even under the Aggressive preset with spatial grid overlays enabled, match confidence scores dropped by less than 0.02 cosine similarity units.

6.2 Why Varman 2.0 Failed Completely

Detailed post-mortem analysis revealed why analytical CPU image filtering cannot defend against deep face recognition:

1. **Representation Invariance:** Modern deep face recognition architectures (ResNets, Inception networks) are trained with extensive data augmentation, including random affine transformations, elastic warping, blur, and color jitter. They map faces to identity manifolds based on deep semantic features (relative eye-to-nose geometry, receptive field embeddings) that are completely robust against $1.5%$ coordinate warping.
2. **Sub-manifold Manifold Projection:** The mid-frequency DCT noise injected in Stage 3 was spatially uniform and uncorrelated with the loss gradients of the target models. In high-dimensional embedding space, random or uniform perturbations project onto orthogonal subspaces, leaving the principal identity components unchanged.
3. **The Guardrail Paradox:** The SSIM quality guardrail actively sabotaged any potential protective effect. Whenever warping or DCT noise began to push pixel values enough to potentially alter neural network activations, the SSIM guardrail attenuated the perturbation by $0.85\times$ to protect structural fidelity, guaranteeing that the image remained within the model's clean classification manifold.

---

7. Retraction of Historical Claims and Legacy Transition

The empirical finding that Varman 2.0 provided zero protection necessitated an immediate, transparent retraction of earlier project claims:

- **Retired Headline:** The claim that "*Varman provides a real-time, non-iterative AI face cloak achieving 93% disruption*" was formally retracted. As recorded in Handover Section 3 (lines 64–66 [Measured]):

  > *"The old 'ArcFace 0.41 / FaceNet 0.44 / 93% disruption' numbers are not results (0.4427 is FaceNet's 1%-FAR threshold; 0.41 was a design target). Never use them."*

- **Engineering Reclassification:** Rather than deleting Varman 2.0, the codebase was refactored on October 9, 2026 into a dual-mode architecture:
  - `mode="fast"` : Preserves the Varman 2.0 CPU pipeline, honestly documented as a non-protective image enhancement and landmark warping filter.
  - `mode="strong"` : Initiated as a greenfield research branch (`adaptive-strength`) dedicated to developing true mathematical adversarial protection via canonical GPU-accelerated EOT-PGD.

Part IV: Varman 3.0 (`adaptive-strength`) — Canonical Aligned EOT-PGD

1. Mathematical Formulation and Architectural Foundation

On October 9, 2026, Project Varman initiated its third and definitive architectural phase on the `adaptive-strength` git branch (commit `11f7894` [Measured]). Having proven that global image filtering (v2.0) provided zero protection against deep facial recognition manifolds, Varman 3.0 returned to first principles: **gradient-based adversarial perturbation via Expectation Over Transformation (EOT) Projected Gradient Descent (PGD)**.

However, learning from the fatal downsampling collapse of AMOR 1.0, Varman 3.0 made a critical architectural breakthrough: **the perturbation was formulated and optimized directly within canonical 112×112 aligned facial space**, bypassing full-canvas image resizing entirely.

```
flowchart TD
    Orig["Original Input Image (H x W)"] --> Det["YuNet 5-Point Landmark Detection"]
    Det -->|Affine Matrix M| Crop["Canonical Aligned Crop x_clean (112x112)"]
    Crop --> Loop["PGD Optimization Loop (25 Iterations)"]

    subgraph EOT_Loop ["Inside Differentiable Optimization Loop"]
        Pert["Current Perturbation delta"] --> Adv["x_adv = clamp(x_clean + delta)"]
        Adv --> InvWarp["Inverse Affine Warp to Canvas (torch.grid_sample)"]
        InvWarp --> Blend["Elliptical Feathered Blending with Full Canvas"]
        Blend --> DiffJPEG["Differentiable JPEG Compression (Q in [50, 95])"]
        DiffJPEG --> ReAlign["Differentiable Forward Re-Alignment to 112x112"]
        ReAlign --> Surrogates["Dual Surrogates (ArcFace IResNet50 + FaceNet InceptionResnetV1)"]
        Surrogates --> Loss["Loss = sim(ArcFace) + 1.0 * sim(FaceNet)"]
        Loss --> Backward["Autograd Backward: dLoss / d_delta"]
        Backward --> Momentum["Update Momentum (mu = 0.90) & Step (alpha = eps / 7)"]
        Momentum --> Proj["Luminance-Only Projection (chroma_scale = 0.0) & Clip to [-eps, eps]"]
    end

    Loop --> EOT_Loop
    EOT_Loop --> FinalPert["Final Perturbation delta*"]
    FinalPert --> RealComp["Real OpenCV warpAffine Feathered Composite"]
    RealComp --> Output["Final Protected Image Output"]
```

1.1 Canonical Alignment Invariant

Modern deep facial recognition models (including ArcFace, SFace, and MobileFaceNet) do not process raw photographs directly. During inference, they detect facial landmarks and apply a 5-point similarity transformation to extract an aligned 112×112 RGB facial crop.

- By defining the adversarial perturbation tensor $\delta \in \mathbb{R}^{1 \times 3 \times 112 \times 112}$ strictly on this canonical crop, Varman 3.0 ensured that every perturbed pixel directly maps to the input receptive field of the target neural networks.
- Upsampling and downsampling filters no longer wash out the noise, because the perturbation is natively aligned with the facial feature detectors.

1.2 Dual Gradient Surrogates

Varman 3.0 employs dual white-box backbones implemented in `app/protection/surrogate_models.py` [Measured]:

1. **ArcFace Backbone:** An IResNet50 architecture loaded from `~/.cache/varman/arcface_r50_backbone.pth` (174.68 MB, SHA-256 `ef005ba502...` [Measured]), pretrained on MS1MV3. Extracts 512-dimensional L_2-normalized embeddings.
2. **FaceNet Backbone:** An InceptionResnetV1 architecture loaded via `facenet_pytorch` (`20180402-114759-vggface2.pt`, 106.71 MB, SHA-256 `281cebca86...` [Measured]), pretrained on VGGFace2. Inputs are bilinearly interpolated from 112×112 to 160×160 before embedding extraction.

The combined adversarial loss minimizes cosine similarity between protected and clean embeddings: $$\mathcal{L}_{adv} = \cos(f_{AF}(x_{adv}), f_{AF}(x_{clean})) + w_{fn} \cdot \cos(f_{FN}(x_{adv}), f_{FN}(x_{clean}))$$ where $w_{fn} = 1.0$ is the empirically validated surrogate weight.

1.3 Momentum Iterative PGD (MI-FGSM)

To navigate the complex, non-convex multi-surrogate loss landscape, the engine accumulates normalized gradient velocity over $N=25$ steps: $$g_{t+1} = \mu \cdot g_t + \frac{\nabla_\delta \mathcal{L}(\delta_t)}{|\nabla_\delta \mathcal{L}(\delta_t)|*1}$$ $$*delta*{t+1} = \operatorname{Proj}*{*mathcal{S}}*left(*delta_t - *alpha *cdot *operatorname{sgn}*{g*{t+1}})\right)$$ with momentum decay $\mu = 0.90$ and step size $\alpha = \varepsilon / 7.0$ [Measured].

1.4 Differentiable Frequency-Domain JPEG (`DiffJPEG`)

To guarantee survivability against lossy JPEG compression, Varman incorporates a fully differentiable JPEG module (`app/protection/diff_jpeg.py` [Measured]):

- Executes exact 8×8 block 2D DCT and quantization table scaling.
- Quantization rounding is implemented via the **Straight-Through Estimator (STE)**: $$x_\text{quant} = x + \operatorname{detach}(\operatorname{round}(x) - x)$$ allowing continuous gradient backpropagation through integer rounding.
- In each EOT step, the JPEG quality factor is sampled uniformly: $Q \sim \mathcal{U}(50, 95)$ [Measured].

1.5 Luminance-Only Projection

Color distortions in RGB space frequently trigger noticeable chromatic blotches. Varman converts RGB delta tensors into luminance (Y) and chrominance components: $$d_Y = \frac{1}{3}(d_R + d_G + d_B)$$ $$d_\text{chroma} = \delta - d_Y$$ $$\delta_\text{projected} = d_Y + \text{chroma_scale} \cdot d_\text{chroma}$$ Setting $\text{chroma_scale} = 0.0$ forces perturbations to exist strictly on the luminance channel, reducing CIEDE2000 color difference by ~40% at identical epsilon budgets [Measured].

---

2. Chronological Benchmark Campaign

The development of Varman 3.0 proceeded through an exhaustive, hypothesis-driven experimental sequence where every claim was validated on physical saved-and-reloaded artifacts.

2.1 Step 0: Testbed Hardening and Calibration Split (Oct 9–10, 2026)

In commits `11f7894` and `63d8aa9` [Measured], the evaluation suite was established:

- **Dataset:** 150 LFW genuine pairs ($N=150$) evaluated across 4 models: FaceNet, ArcFace, SFace, and MobileFaceNet.
- **Threshold Calibration:** To eliminate ungrounded similarity claims, thresholds were strictly calibrated on an independent split of **1,500 LFW impostor pairs** (`calibrated_thresholds_sweep.json` [Measured]).
  - **1% FAR Thresholds:** FaceNet **`0.3754`**, ArcFace **`0.1476`**, SFace **`0.2920`**, MobileFaceNet **`0.1688`** [Measured].
  - **0.1% FAR Thresholds:** FaceNet **`0.4555`**, ArcFace **`0.1828`**, SFace **`0.3315`**, MobileFaceNet **`0.2065`** [Measured].
- **Valid Denominators:** Biometric protection requires genuine pairs to match when clean. The valid baseline match counts across 150 pairs at 1% FAR were established as:
  - FaceNet: **143 / 150**
  - ArcFace: **142 / 150**
  - SFace: **141 / 150**
  - MobileFaceNet: **142 / 150** [Measured]. At 142 valid pairs, each individual pair flipping state accounts for approximately **`0.70%`** per pair per model [Measured].

2.2 Epsilon Calibration and Real-Photo Verification (Oct 11, 2026)

In commit `d4e52be` [Measured], the initial epsilon sweep was evaluated on LFW and on 8 real portrait photos (26 directed pairs) using crop-only PGD:

Epsilon Budget	LFW 4-Model Mean Flip (1% FAR PNG)	LFW Mean Flip (1% FAR Q75)	Real Photos (1% FAR PNG)	Face SSIM (LFW)	CIEDE Mean (LFW)
$\varepsilon = 6/255$	13.4% [M]	11.4% [M]	14.4% [M]	0.967 [M]	1.09 [M]
$\varepsilon = 7/255$	14.4% [M]	13.2% [M]	23.1% [M]	0.958 [M]	1.23 [M]
$\varepsilon = 8/255$	21.6% [M]	21.5% [M]	25.0% [M]	0.949 [M]	1.36 [M]

Reading: $\varepsilon=8/255$ established the first viable baseline (~21.6% LFW protection). Real photos exhibited higher flip rates (25.0%), proving that crop-based PGD was functioning on unconstrained consumer cameras [Measured].

2.3 PGD Step Convergence Benchmark (Oct 12, 2026)

Commit `490e8b3` (`benchmarks/results/pgd_convergence_q50_exp/` [Measured]) evaluated whether increasing optimization iterations from 25 to 50 or 75 steps could bridge the gap to the 40% project target.

- **Condition A (25 steps fixed):** 22.00% (PNG) / 18.30% (Q50) [Measured]. Runtime: 2.08 s/img.
- **Condition B (50 steps fixed):** 24.10% (PNG) / 20.41% (Q50) [Measured]. Runtime: 4.09 s/img.
- **Condition C (75 steps fixed):** 23.23% (PNG) / 18.30% (Q50) [Measured].
- **Condition D (50 steps cosine):** 24.45% (PNG) / 19.01% (Q50) [Measured].
- **Condition E (75 steps cosine):** 23.93% (PNG) / 19.18% (Q50) [Measured].

Conclusion: Doubling or tripling PGD steps produced marginal gains (+1 to +2 pp at PNG) and zero improvement under real JPEG Q50 (18.30% vs 18.30% [Measured]), while doubling computation time. **25 steps was locked as the optimal Pareto operating point.**

2.4 The Threshold-Aimed Hinge Loss Experiment (Oct 13, 2026)

Commit `feeb23d` (`run_threshold_loss_experiment.py` [Measured]) tested a custom hinge loss designed to halt optimization once a surrogate's embedding crossed its 1% FAR threshold plus a 15% safety margin:

- **Baseline Cosine Loss (A rerun):** 22.70% (PNG) / 20.24% (Q75) / 17.95% (Q50) [Measured].
- **Hinge Loss (15% Margin):** 17.93% (PNG) / 17.57% (Q75) / 17.75% (Q50) [Measured].
- *Failure Mechanism:* Because ArcFace was easier to disrupt than FaceNet, ArcFace reached its hinge margin early in the optimization run. Its gradient immediately dropped to zero, leaving all remaining gradient steps to optimize FaceNet alone. Consequently, ArcFace protection collapsed, reducing overall multi-model protection. Hinge loss was permanently abandoned.

2.5 Surrogate Loss Weight Ratio Experiments (Oct 13, 2026)

Commits `b20de80` and `22c02c0` systematically ablated FaceNet's loss weight w_fn:

- $w_\text{fn} = 0.10$: 14.62% PNG / 11.98% Q50 [Measured].
- $w_\text{fn} = 0.25$: 18.66% PNG / 17.25% Q50 [Measured].
- $w_\text{fn} = 0.50$ (Default): 22.00% PNG / 17.25% Q50 [Measured].
- $w_\text{fn} = 1.00$: **25.50% PNG / 21.28% Q50** [Measured].
- $w_\text{fn} = 2.00$: 24.60% PNG / 22.49% Q50 [Measured].

Outcome: Setting $w_\text{fn} = 1.0$ (equal weighting between ArcFace and FaceNet) delivered the highest multi-model protection (+3.5 pp 4-model mean, +3.2 pp held-out at Q50 [Measured]) and was adopted across all subsequent benchmarks.

---

3. The 40-Pair Pipeline Stage Diagnostic & The Paste-Back Rebound

On October 13, 2026, the author executed a granular stage diagnostic on 40 LFW pairs to trace exactly where adversarial disruption was being lost in the post-optimizer pipeline (`22c02c0` , Handover Section 6.4 [Measured]):

Pipeline Evaluation Stage	ArcFace Mean Similarity	ArcFace Below Threshold (%)	FaceNet Mean Similarity	FaceNet Below Threshold (%)
Unprotected Clean Baseline	+0.7038	0/40 (0.0%) [M]	+0.7681	0/40 (0.0%) [M]
Stage 1: Inside Optimizer (112x112 Crop)	**-0.2078**	**40/40 (100.0%)** [M]	**+0.1241**	**35/40 (87.5%)** [M]
Stage 2: Pasted into Full Image (Float32)	-0.0288	36/40 (90.0%) [M]	+0.3005	24/40 (60.0%) [M]
Stage 3: 8-bit PNG Saved & Reloaded	+0.0505	28/40 (70.0%) [M]	+0.3741	19/40 (47.5%) [M]
Stage 4: Real JPEG Q50 Saved & Reloaded	+0.1470	19/40 (47.5%) [M]	+0.4116	17/40 (42.5%) [M]

The Breakthrough Finding

The stage diagnostic revealed the central bottleneck of the entire project:

1. **The optimizer was performing flawlessly:** Inside the 112×112 optimization loop, ArcFace was disrupted on **100% of pairs** and FaceNet on **87.5% of pairs** [Measured].
2. **The collapse occurred post-optimization:** When the perturbed crop was blended back into the original photo using OpenCV's inverse warp, feathered at the edges, quantized to 8-bit PNG, and re-aligned by the evaluator, **over half of the adversarial disruption vanished**.
3. *Root Cause:* The optimizer was optimizing a standalone crop that never experienced inverse warping, canvas blending, or re-alignment. The solution was obvious: **the optimizer must see the entire compositing pipeline during backpropagation**.

---

4. Differentiable Paste-Aware Optimization (Oct 14, 2026)

On October 14, 2026, the team engineered `benchmarks/run_paste_aware_experiment.py` (commit `7268342` [Measured]). Instead of optimizing the crop in isolation, the forward pass was made end-to-end differentiable using PyTorch grid sampling:

1. δ was added to x_clean.
2. The perturbed crop was warped into full-canvas coordinates via `torch.nn.functional.grid_sample()` using the inverse affine matrix M^{-1}.
3. The warped face was blended with the original unperturbed background using an elliptical Gaussian mask.
4. The full composite image was subjected to DiffJPEG compression.
5. The full image was re-aligned back to 112×112 using forward grid sampling.
6. Gradients were backpropagated through the entire pipeline into δ.

4.1 Empirical Results (150 LFW Pairs, 1% FAR)

The paste-aware engine delivered a massive, unprecedented performance leap:

Experimental Condition	PNG 4-Model Mean / Held-Out	JPEG Q75 Mean / Held-Out	JPEG Q50 Mean / Held-Out
Row 0: Control	0.00% / 0.00% [M]	0.35% / 0.35% [M]	0.17% / 0.00% [M]
Row 1: Crop-Only ($w=1.0$)	25.85% / 22.27% [M]	23.38% / 20.50% [M]	21.10% / 19.45% [M]
Row 2: Paste-Aware ($w=1.0$)	**33.57% / 25.09%** [M]	**32.16% / 24.38%** [M]	**30.22% / 23.68%** [M]
Row 3: Paste-Aware + Landmark Jitter	**40.09% / 33.92%** [M]	**39.57% / 32.87%** [M]	**36.57% / 30.74%** [M]

Key Findings:

- **Row 2 (Paste-Aware):** Surged Q50 protection by **+9.12 percentage points** over crop-only (30.22% vs 21.10% [Measured]). Paired exact McNemar tests confirmed statistically significant gains on ArcFace ($p=0.0192$) and FaceNet ($p=4.18 \times 10^{-7}$) [Measured].
- **Row 3 (Paste-Aware + Jitter):** Injecting ± 2 px translation and $\pm 3%$ scaling jitter into the alignment loop pushed Q50 protection to **36.57%** 4-model mean and **30.74%** held-out mean [Measured], approaching the owner's 40% success bar for the first time.

---

5. Landmark Drift and The K82E1L Viewing Audit (Oct 15, 2026)

5.1 YuNet Landmark Drift Controlled Pilot (Commit `14e61f0`)

A standing hypothesis was that adversarial noise might cause YuNet to detect shifted facial landmarks during re-detection, destroying alignment.

- To test this, the pilot directly measured landmark Euclidean coordinates on 30 pairs before and after protection (`LANDMARK_DRIFT_PILOT_REPORT_2026-10-15.md` [Measured]).
- **Measured Displacement:** Mean drift was **1.33 px** (R1), **1.45 px** (R2), and **1.49 px** (R3) [Measured]. The 95th percentile drift was only 2.8 px, and face detection was 100% maintained.
- *Conclusion:* Landmark drift was negligible and already fully covered by the optimizer's ± 2 px jitter. The drift-collapse hypothesis was formally refuted.

5.2 K82E1L Human Viewing Audit & Budget Rejection

An audit of an earlier human viewing test revealed that an evaluator preferred a specific protected image (`K82E1L_SBS`) generated under an adaptive bisection search.

- When `K82E1L` was benchmarked, it exhibited **zero protection** ($0/8$ flips across all models and formats) because the bisection search had attenuated the perturbation to $s = 0.2891$ ($\varepsilon \approx 2.3/255$) [Measured].
- Attempting to enforce the evaluator's aesthetic budget (Face SSIM ≥ 0.96, CIEDE Mean ≤ 1.10) rejected every working PGD configuration (`K82E1L_CONFIG_AND_BUDGET_2026-10-15.md` [Measured]). The budget was formally rejected as unviable for biometric defense.

---

6. Region-Size Ablation and Metric Reconciliation (Oct 15, 2026)

6.1 Region-Size Ablation Benchmark (Commit `92ac31e`)

Having established that facial boundaries play a critical role in feature extraction, commit `92ac31e` evaluated mask scale expansion:

- **Scale 0.70 (Tight Mask, 33.87% coverage):** Q50 protection collapsed to **17.20%** 4-model mean [Measured].
- **Scale 1.00 (Standard Mask, 64.52% coverage):** Achieved **37.10%** Q50 protection [Measured].
- **Scale 1.30 (Expanded Mask, 92.11% coverage):** Pushed Q50 protection to **48.36%** 4-model mean and **40.64%** held-out mean [Measured].

6.2 The Face SSIM Metric Discrepancy & Reconciliation

The region ablation benchmark uncovered a severe numerical contradiction in reported metrics:

- `run_paste_aware_experiment.py` (commit `7268342`) reported Face SSIM **`0.9427`** for $\varepsilon=8$, scale 1.0.
- `run_region_ablation.py` (commit `92ac31e`) reported Face SSIM **`0.8753`** for identical parameters.

```
flowchart LR
    subgraph DefA ["Definition A (Inflated)"]
        A1["Re-extracted 112x112 Composite"] --> A2["Evaluate ALL 12,544 Pixels"]
        A2 --> A3["Unperturbed feathered corners inflate SSIM to ~0.9414"]
    end
    subgraph DefC ["Definition C (Pessimistic)"]
        C1["Raw PGD Crop Before Pasting"] --> C2["Evaluate Raw 112x112 Tensor"]
        C2 --> C3["Omits feathered blending; drops SSIM to ~0.8753"]
    end
    subgraph DefB ["Definition B (Standardized Standard)"]
        B1["Re-extracted 112x112 Composite"] --> B2["Apply Canonical Face Mask (>= 0.10)"]
        B2 --> B3["Evaluates strictly on facial anatomy; baseline SSIM = 0.9192"]
    end
```

The discrepancy was traced to exact code implementations:

1. **Definition A (Inflated):** Computed SSIM over the entire 112×112 re-extracted composite crop. Because 25–35% of the outer canvas contained unperturbed clean background pixels feathered into the image, SSIM was artificially inflated.
2. **Definition C (Pessimistic):** Computed SSIM on the raw PGD crop before compositing, omitting the feathered blend entirely.
3. **Definition B (Standardized):** Evaluates SSIM strictly within the fixed canonical face mask (≥ 0.10, covering 8,095 pixels) on the aligned 112 crop, compared against the uncompressed source. Definition B measures true facial distortion independently of mask scaling and was formally standardized across all subsequent research (`metric_reconciliation.json` [Measured]).

---

7. The Coverage Frontier Benchmark: F0 Through F5 (Oct 15, 2026)

Commit `7eaf0e5` (`COVERAGE_FRONTIER_REPORT_2026-10-15.md` [Measured]) executed the comprehensive coverage frontier across 150 LFW genuine pairs, evaluating six distinct conditions under standardized Definition B:

Row ID	Config (ε, scale)	Face SSIM (Def B)	CIEDE Mean	CIEDE p95	Real JPEG Q50 4-Model Mean (1% FAR)	Real JPEG Q50 Held-Out Mean (1% FAR)
F0 (Baseline)	$8/255, 1.00$	**0.9192** [M]	**1.5037** [M]	3.0674 [M]	**36.39%** [M]	**30.39%** [M]
F1 (Max Protection)	$8/255, 1.30$	**0.8944** [M]	**1.7722** [M]	3.2276 [M]	**49.95%** [M]	**42.41%** [M]
F2 (Matched Candidate)	$7/255, 1.30$	**0.9129** [M]	**1.5952** [M]	**2.9206** [M]	**37.10%** [M]	**30.39%** [M]
F3	$6/255, 1.30$	**0.9308** [M]	**1.4138** [M]	2.6236 [M]	**26.55%** [M]	**21.57%** [M]
F4	$5/255, 1.30$	**0.9471** [M]	**1.2378** [M]	2.3314 [M]	**13.90%** [M]	**13.43%** [M]
F5 (Intermediate)	$8/255, 1.15$	**0.8986** [M]	**1.7418** [M]	3.1885 [M]	**45.73%** [M]	**39.23%** [M]

7.1 Pre-Registered Scientific Findings

- **Q1 (Replication):** Row `F1` replicated the region ablation findings within ~1.5 pp (49.95% vs 48.36% [Measured]), confirming experimental stability.
- **Q2 (Dose-Response Across Scales):** A monotonic progression was proven: $\text{F0 } (36.39%) < \text{F5 } (45.73%) < \text{F1 } (49.95%)$ [Measured]. Paired McNemar tests proved F5 significantly outperformed F0 across all four models ($p \le 0.0074$ after Holm correction) [Measured].
- **Q3 (Matched Protection Candidate):** Row `F2` ($\varepsilon=7$, scale 1.30) matched the protection of baseline F0 (37.10% vs 36.39% 4-model; McNemar $p \ge 0.72$ on all models [Measured]) while achieving superior peak color fidelity (CIEDE p95 **`2.9206`** vs `3.0674` [Measured]).
- **Q4 (The Steep Epsilon Frontier):** At scale 1.30, reducing epsilon from 8 down to 5 produced a steep, linear collapse in protection (~11–13 pp per epsilon step: $49.95% \to 37.10% \to 26.55% \to 13.90%$ [Measured]).

7.2 Graded Grain Smoothing Measurement

Step 3 of the frontier benchmark synthesized phase-randomized grain with radial falloff into the background outside the facial boundary (`graded_grain/` [Measured]):

- Evaluated on 50 pairs of F1, graded grain maintained 100% YuNet detection and low landmark displacement (2.18–2.21 px [Measured]).
- High-pass boundary conspicuity proxy ratio dropped from **20.19** (F1 baseline) to **8.61** (amplitude 0.25) and **4.44** (amplitude 0.50) [Measured], smoothing the hard boundary step with only a modest impact on Q50 protection (42.5% vs 47.0% [Measured]).

---

8. Human Review Campaign and The Arithmetic Audit (Oct 16, 2026)

8.1 50-Stimulus Quick Review (`quick_review_10groups`)

To obtain empirical human feedback without overwhelming the rater, 10 complete source groups (50 stimuli total: F0, F1, F2, Unprotected, F1 Graded amp 0.25) were deterministically sampled using seed `20261015` (`OR_001` to `OR_044` , pair indices 0, 2, 3, 13, 14, 30, 31, 33, 36, 48) and rated by the evaluator on a 0–2 scale (0=None, 1=Slight, 2=Clear):

- **Unprotected Control:** 10/10 rated `0` (None), mean `0.00` [Measured]. Confirmed evaluator calibration and zero baseline false alarms.
- **F0 ($\varepsilon=8$, scale 1.00):** 7 Slight, 3 Clear, mean `1.30` [Measured].
- **F2 ($\varepsilon=7$, scale 1.30):** 3 Slight, 7 Clear, mean `1.70` [Measured].
- **F1 ($\varepsilon=8$, scale 1.30):** 10 Clear, mean `2.00` [Measured].
- **F1 Graded Grain (amp 0.25):** 10 Clear, mean `2.00` [Measured].

8.2 Audit of the F0 vs. F2 Paired Arithmetic Error

An inconsistency was detected in earlier agent reporting: a narrative summary claimed F0 was cleaner than F2 in 4 groups, tied in 6, and F2 cleaner in 0.

- **Reconciliation from Source Records:** Auditing raw CSV files (`source_group_comparison.csv` and `joined_ratings.csv` [Measured]) established the correct counts:
  - F0 rated cleaner than F2: **5 groups** (`SG_002` , `SG_013` , `SG_028` , `SG_031` , `SG_044`) [Measured].
  - F2 rated cleaner than F0: **1 group** (`SG_003`) [Measured].
  - Tied: **4 groups** (`SG_001` , `SG_014` , `SG_029` , `SG_034`) [Measured].
- The discrepancy was traced to an agent synthesis hallucination that misread group 03. The underlying condition means (F0 1.30 vs F2 1.70) remained mathematically accurate.

8.3 F5 Add-on Visual Screening

To test whether intermediate candidate F5 ($\varepsilon=8$, scale 1.15) could provide a superior compromise, an isolated review package was created (`f5_addon_review/` [Measured]) displaying F5 against clean references across the same 10 source groups:

- **Ratings Distribution:** 0 None, 5 Slight, 5 Clear; mean **`1.50`** [Measured].
- **Paired with F0:** F5 tied F0 in 8 groups and was more conspicuous in 2 groups (`SG_013` , `SG_031`), while delivering a **+9.34 percentage point** protection surge at Q50 (45.73% vs 36.39% [Measured]).
- **Paired with F2:** F5 was cleaner in 3 groups (`SG_002` , `SG_028` , `SG_044`), tied in 6 groups (`SG_001` , `SG_013` , `SG_014` , `SG_029` , `SG_031` , `SG_034`), and more conspicuous in 1 group (`SG_003`).
- **Decoupling of Face SSIM from Perceived Conspicuity:** The screen proved that objective Face SSIM does not predict human conspicuity across groups: F5 groups rated `1` (Slight) had a mean Face SSIM of **`0.88323`** , while groups rated `2` (Clear) had a virtually identical mean Face SSIM of **`0.88438`** [Measured].

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9. Multi-Quality Intermediate JPEG Validation (Oct 16, 2026)

The final empirical benchmark re-encoded pre-JPEG outputs across six distinct quality tiers ($Q \in {95, 90, 75, 70, 60, 50}$) across all 150 LFW genuine pairs (`intermediate_jpeg_results.json` [Measured]):

```
xychart-beta
    title "Figure: 4-Model Protection vs. JPEG Quality Q (150 LFW Pairs)"
    x-axis [Q50, Q60, Q70, Q75, Q90, Q95]
    y-axis "4-Model Protection Rate (%)" 0 --> 60
    line [36.39, 39.57, 39.56, 39.74, 42.03, 40.62]
    line [45.73, 45.90, 45.73, 46.96, 47.84, 48.54]
    line [49.95, 51.37, 51.36, 52.59, 55.59, 54.53]
```

9.1 Validated Protection Monotonicity

1. **Quality Progression:** Protection peaks at high qualities (Q90–Q95) where high-frequency adversarial gradients are minimally quantized, and degrades smoothly toward Q50 (F0: $42.03% \to 36.39%$; F5: $48.54% \to 45.73%$; F1: $55.59% \to 49.95%$ [Measured]).
2. **Control Stability:** Unprotected control false alarms remained negligible across all qualities ($\le 0.35%$ [Measured]).
3. **Sampling Noise:** Differences of 1–2 percentage points across adjacent quality tiers (e.g., F5 scoring 48.54% at Q95 vs 47.84% at Q90) are within standard binomial sampling noise ($\approx 0.70%$ per pair per model).

On October 16, 2026, preset `F5` was integrated into production code (`c5a20a2` [Measured]), establishing the permanent final state of Project Varman.

Part V: Cross-Cutting Engineering & Methodological Analysis

1. Inventory of Retired Headline Claims

Throughout the six-month evolution of Project AMOR and Project Varman, multiple technical claims and headline metrics were published, benchmarked, and subsequently retired or retracted. A core commitment of this retrospective is complete transparency regarding these claims.

Retired Headline Claim	Original Assertion / Source	Measured Reality / Replacement Finding	Root Cause / Reason for Retirement
"93% Disruption Rate / ArcFace 0.41 / FaceNet 0.44"	Claimed in early AMOR/Varman prototypes as proof of evasion.	**False metric.** 0.4427 is FaceNet's 1% FAR threshold on LFW; 0.41 was an arbitrary design goal.	Cosine similarity below 0.44 does not mean 93% disruption. Replaced by empirical 1% FAR flip counting over clean matches [M].
"Real-Time Non-Iterative AI Face Cloak (Varman 2.0)"	Claimed in August 2026 that CPU landmark warping and DCT noise protected photos in <150 ms.	**0.0% protection.** Tested against 4 deep models across 150 pairs, exactly zero identities were disrupted [M].	Deep neural network embeddings are invariant to mild spatial warping and mid-frequency DCT noise. Reclassified as non-protective `mode="fast"`.
"Imperceptible / Invisible Adversarial Defense"	Claimed that adversarial noise was completely imperceptible to human observers.	**Noticeable artifacts.** F1 received 10/10 Clear (mean 2.00); F0 received 7 Slight / 3 Clear (mean 1.30); F5 received 5 Slight / 5 Clear (mean 1.50) [M].	Evasion-capable perturbations induce visible boundary steps and high-frequency luminance noise on smooth facial skin.
"Y-NPR Proves Robust Frequency Survival"	Claimed Y-NPR scores of 72–88% mathematically demonstrated JPEG survival.	**Flawed proxy metric.** Y-NPR was hard-clamped at 100% in code and does not correlate with feature evasion.	High-frequency luminance energy can survive JPEG quantization without disrupting deep neural network feature maps. Retired permanently.
"Face SSIM Predicts Visual Naturalness"	Assumed higher Face SSIM values indicate superior human aesthetic preference.	**Empirical decoupling.** In F5, groups rated 1 had mean SSIM `0.88323`, while groups rated 2 had mean SSIM `0.88438` [M].	SSIM measures pixel luminance/contrast variance, which cannot differentiate uniform texture noise from conspicuous boundary steps.
"YuNet Landmark Drift Collapsed the Defense"	Hypothesized that adversarial noise shifted landmark coordinates, destroying alignment.	**Disproven by direct measurement.** Mean landmark drift was only 1.33–1.49 px (p95 ~2.8 px) [M].	Landmark displacement was negligible and fully covered by the optimizer's ± 2 px spatial jitter (`14e61f0`).
"The Perturbation Epsilon Budget is the Primary Bottleneck"	Assumed that increasing ε was required to overcome low protection rates.	**Disproven by stage diagnostic.** 50–60% of disruption was being destroyed by post-optimizer compositing [M].	The bottleneck was the forward pass architecture, resolved by differentiable paste-aware optimization (`7268342`).
"Protection Ceiling is ~50%"	Broadly characterized the upper bound as roughly 50%.	**Refined to roughly 50–56%.** F1 achieves 49.95% at Q50, rising to 55.59% at Q90 across 4 models at 1% FAR [M].	Updated across all reports to accurately capture the Q50 to Q90 range under the strongest configuration.

2. Catalog of Negative Results, Empirical Rejections, and Dead Ends

Science progresses by identifying and cataloging failures. The following technical approaches were thoroughly implemented, benchmarked, and formally rejected as dead ends:

```
mindmap
  root((Varman Dead Ends))
    Full-Canvas PGD
      512x512 canvas
      Lanczos blur washing
      100% failure rate
    Heuristic CPU Filtering
      YuNet landmark warping
      Luminance DCT noise
      0% black-box protection
    PGD Step Scaling
      50 and 75 steps
      Zero Q50 gain
      2x runtime cost
    Loss Modifications
      Hinge threshold loss
      ArcFace gradient starvation
      -4.77 pp drop
    Weight Reductions
      FaceNet weight under 0.5
      Protection collapse to 14%
    Aesthetic Budget Clamping
      K82E1L budget s=0.2891
      0 of 8 flips
      Rejects all working PGD
    Spatial Heuristics
      Texture weighting Sobel
      Spatial grid sinusoidal
      Graded grain halo smoothing
```

2.1 Dead End 1: Full-Canvas 512×512 Optimization (AMOR 1.0)

- **Concept:** Optimize the entire letterboxed image tensor against deep face recognition models.
- **Outcome: 100% failure rate** [Measured].
- **Why it Failed:** Resampling high-resolution images to 512×512 and upsampling back via Lanczos interpolation acted as a low-pass filter, completely destroying high-frequency adversarial noise. Background pixels absorbed gradients, causing massive visual degradation with zero facial protection.

2.2 Dead End 2: Non-Iterative CPU Landmark Warping & DCT Filtering (Varman 2.0)

- **Concept:** Apply subtle spatial warping to facial landmarks and mid-frequency luminance DCT noise to bypass iterative GPU optimization.
- **Outcome: 0.0% protection** against modern deep face recognition models [Measured].
- **Why it Failed:** Deep convolutional networks and vision transformers are explicitly trained to be invariant to mild spatial deformations and random frequency perturbations. The heuristic filters acted as benign data augmentations.

2.3 Dead End 3: Scaling PGD Steps Beyond 25 (Commit `490e8b3`)

- **Concept:** Increase PGD iterations from 25 to 50 or 75 steps, using fixed and cosine learning rate schedules, to find deeper adversarial minima.
- **Outcome:** Marginal gains on uncompressed PNG (+1 to +2 pp), but **0.0 pp gain under real JPEG Q50** (18.30% vs 18.30% [Measured]), while doubling latency (4.09 s vs 2.08 s/image [Measured]).
- **Why it Failed:** Additional iterations overfitted high-frequency decision boundaries of the surrogate models that are immediately eliminated by JPEG DCT quantization tables.

2.4 Dead End 4: Threshold-Aimed Hinge Loss (Commit `feeb23d`)

- **Concept:** Replace unbounded cosine loss with a margin hinge loss that ceases gradient backpropagation once similarity drops below the 1% FAR threshold plus a 15% safety margin.
- **Outcome:** Overall 4-model protection **dropped from 22.70% to 17.93%** [Measured].
- **Why it Failed:** ArcFace similarity dropped below its target threshold within the first 8–10 iterations. Its gradient switched off completely, leaving remaining steps to optimize FaceNet alone. This gradient starvation allowed ArcFace protection to collapse.

2.5 Dead End 5: Suppressing FaceNet Surrogate Weight Below 0.5 (Commit `b20de80`)

- **Concept:** Reduce FaceNet loss weight ($w_\text{fn} \in {0.10, 0.25}$) to focus optimizer capacity on ArcFace transferability.
- **Outcome:** Severe protection collapse: 4-model mean dropped to **14.62%** at $w=0.10$ and **18.66%** at $w=0.25$ [Measured].
- **Why it Failed:** FaceNet's Inception architecture provides crucial multi-scale spatial gradients. Suppressing it produced perturbations that failed to generalize across architectures.

2.6 Dead End 6: Enforcing the K82E1L Evaluator Aesthetic Budget

- **Concept:** Constrain PGD perturbation using an evaluator-preferred aesthetic budget (Face SSIM ≥ 0.96, CIEDE Mean ≤ 1.10).
- **Outcome: Zero protection** ($0/8$ flips across all models and formats [Measured]).
- **Why it Failed:** The evaluator's preferred image had been attenuated by bisection search to $s = 0.2891$ ($\varepsilon \approx 2.3/255$). Enforcing this constraint rejected every working PGD configuration, demonstrating that adversarial protection under JPEG compression requires an epsilon budget $\varepsilon \ge 7/255$.

2.7 Dead End 7: Texture-Weighted Spatial Gradient Masking

- **Concept:** Scale perturbation amplitude by local Sobel luminance gradients $w(x, y) \in [0.1, 1.0]$ to hide noise in high-texture areas (hair, eyebrows) while sparing smooth skin.
- **Outcome:** Evasion rates dropped by >8 percentage points.
- **Why it Failed:** Facial recognition embeddings rely heavily on the smooth cheek and forehead contours to establish facial plane geometry. Dampening noise in smooth skin areas allowed the deep networks to extract pristine embeddings.

2.8 Dead End 8: Graded Grain Boundary Smoothing (Commit `7eaf0e5`)

- **Concept:** Synthesize phase-randomized grain with radial falloff outside the face mask to eliminate the sharp boundary step between perturbed face and clean background.
- **Outcome:** Reduced high-pass spatial conspicuity proxy ratio from 20.19 to 8.61 (amp 0.25) [Measured], but in the 50-stimulus human review, **10/10 images were still rated 2 (Clear artifacts)** [Measured].
- **Why it Failed:** While spatial frequency transitions were smoothed, human raters easily detected the broader halo of synthetic noise extending into the background.

---

3. Bug Register & Defect Remediations

During development and auditing, five critical software defects were discovered, diagnosed, and resolved:

3.1 Defect 1: The `facenet-pytorch` METADATA Corrupt Packaging Bug

- **Symptom:** Installing packages in the virtual environment threw parsing errors in `facenet-pytorch==2.6.0` metadata.
- **Diagnosis:** A known upstream issue in `facenet-pytorch` wheels containing corrupted email formatting in its distribution metadata.
- **Remediation:** Manually patched the distribution metadata file in `.venv` and documented the requirement to install via `--no-deps` in future environment bootstraps (Handover Section 5, line 104 [Measured]).

3.2 Defect 2: Differentiable Warp Coordinate Alignment Mismatch

- **Symptom:** During early paste-aware trials, the forward grid sampler produced slight blurring relative to OpenCV's `warpAffine`.
- **Diagnosis:** PyTorch `grid_sample` assumes pixel coordinates normalized to $[-1, 1]$ with pixel centers at $(-1 + 1/N)$, whereas OpenCV's `warpAffine` uses integer pixel centers with half-pixel offsets $(0.5, 0.5)$.
- **Remediation:** Formulated an exact coordinate transformation matrix reconciling OpenCV and PyTorch sampling conventions. Implemented an automated pre-flight equivalence check (`Step 2` in `run_paste_aware_experiment.py`) verifying that mean absolute pixel difference was below 0.08 pixel intensity units across test images before executing benchmarks [Measured].

3.3 Defect 3: Face SSIM Definition A Feathering Inflation Defect

- **Symptom:** Earlier benchmark scripts reported Face SSIM values of 0.9427, while region ablation scripts reported 0.8753 for identical configurations.
- **Diagnosis:** Definition A evaluated SSIM over the entire 112×112 rectangular canvas. Because outer corners were feathered to clean image data, ~30% of the evaluated pixels were completely unperturbed, artificially boosting the structural similarity score.
- **Remediation:** Formally standardized **Definition B** across all evaluation engines, computing SSIM strictly within the fixed canonical face mask (≥ 0.10, covering 8,095 pixels) on the aligned 112 crop, compared against the uncompressed source, eliminating canvas-edge artifacts from scientific reporting.

3.4 Defect 4: F0 vs. F2 Paired Arithmetic Discrepancy in Narrative Reporting

- **Symptom:** A summary paragraph in an analysis report claimed F0 was cleaner than F2 in 4 groups, tied in 6, and F2 cleaner in 0, clashing with the per-group table.
- **Diagnosis:** An agent synthesis hallucination misread Group 03 where F2 was rated 1 (Slight) and F0 was rated 2 (Clear).
- **Remediation:** Re-audited raw CSV records (`source_group_comparison.csv` and `joined_ratings.csv` [Measured]), establishing the verified counts: **F0 cleaner in 5 groups, F2 cleaner in 1 group, tied in 4 groups** [Measured]. Corrected across all permanent handover documents and reports.

3.5 Defect 5: Pytest Non-Daemon Thread Interpreter Shutdown Hang

- **Symptom:** Executing `pytest tests/ -q` printed `29 passed in 13.70s` but hung indefinitely during interpreter shutdown, requiring forced termination.
- **Diagnosis:** Thread diagnostic identified lingering non-daemon background threads spawned by SQLAlchemy connection pooling (`_connection_worker_thread`) and OpenCV threadpools (`ThreadPoolExecutor-0_0`) that blocked the Python process exit handler.
- **Remediation:** Added explicit engine disposal fixtures (`engine.dispose()`) in `conftest.py` , configured `cv2.setNumThreads(1)` during test execution, and implemented hard timeout execution harnesses in all verification tasks.

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4. Key Engineering Principles & Architectural Decisions

The final architecture of Varman is underpinned by five core principles:

1. **Decoupling Evaluation Models from Surrogates:** Adversarial privacy systems cannot be objectively evaluated using the same networks used for gradient generation. Varman strictly reserves **SFace** (OpenCV) and **MobileFaceNet** as non-differentiable held-out targets, ensuring that held-out protection rates reflect true black-box transferability.
2. **Physical Disk Verification Gate:** Synthetic differentiable approximations (such as `DiffJPEG`) are indispensable for gradient backpropagation, but they remain mathematical idealizations. All empirical conclusions must be drawn from physical JPEG files saved to disk, reloaded, and detected via independent computer vision pipelines.
3. **Canonical Spatial Alignment:** Perturbations must be optimized in the coordinate space natively ingested by target neural networks (112×112 canonical aligned facial space). Full-canvas or letterboxed optimization is inherently doomed by downstream resampling filters.
4. **End-to-End Differentiable Compositing:** The optimization forward pass must incorporate the entire post-processing pipeline—inverse affine remapping, feathered mask blending, quantization, and forward re-alignment—to prevent the catastrophic 50% post-paste degradation rebound.
5. **Radical Transparency in Metric Reporting:** All protection statistics must declare the exact biometric operating threshold (1% FAR), the evaluation dataset (150 LFW genuine pairs), the number of valid pairs, and the compression quality factor. Vague claims of "evading scrapers" or "imperceptible cloaking" have no place in scientific software engineering.

Part VI: Final System State & Research Closure

1. Final Production Configuration: Preset `F5`

Project Varman concludes with the formal integration of named preset `F5` into the core production engines (`app/protection/strong_engine.py` and `app/protection/protection_engine.py` , commit `c5a20a2` [Measured]).

Preset `F5` represents the empirically verified Pareto optimal compromise between biometric disruption and human visual conspicuousness. It strikes a deliberate balance: avoiding the high facial recognition match rates of baseline `F0` (which allowed ~63.6% true matches under real JPEG Q50) while eliminating the severe, full-face boundary haloing of candidate `F1` (which scored 2.00 / Clear artifacts across all evaluated human screening images).

1.1 Specification of Preset Invariants

The configuration parameters of preset `F5` are frozen as follows:

Hyperparameter Key	Exact Config Value	Mathematical Definition / Operational Role	Verification Status
`eps`	$8.0 / 255.0 \approx 0.03137$	Maximum L_∞ perturbation bound on canonical 112×112 crop	Verified (`test_f5_preset.py`) [M]
`mask_scale`	1.15	Spatial expansion factor for elliptical Gaussian blending mask	Verified (`test_f5_preset.py`) [M]
`steps`	25	Number of PGD gradient descent iterations	Verified (`test_f5_preset.py`) [M]
`alpha`	$(8.0 / 255.0) / 7.0 \approx 0.00448$	Per-iteration gradient step size ($\approx \varepsilon / 7$)	Verified (`test_f5_preset.py`) [M]
`mu`	0.90	Velocity decay momentum factor for MI-FGSM optimization	Verified (`test_f5_preset.py`) [M]
`w_fn`	1.00	Loss weight for FaceNet surrogate ($\mathcal{L} = \text{sim}*text{AF} + 1.0 \cdot *text{sim}\text{FN}$)	Verified (`test_f5_preset.py`) [M]
`luminance_only`	`True`	Forces perturbation projection strictly onto Y (luminance) channel	Verified (`test_f5_preset.py`) [M]
`chroma_scale`	0.00	Zeroes out chromatic perturbation ($d_\text{chroma} = 0.0$)	Verified (`test_f5_preset.py`) [M]
`enable_guardrail`	`False`	Disables post-hoc attenuation, allowing full adversarial power	Verified (`test_f5_preset.py`) [M]
`feather_sigma`	4.0 px	Standard deviation of Gaussian blur applied to elliptical mask	Verified (`test_f5_preset.py`) [M]
`diff_jpeg_range`	$[50, 95]$	Uniform quality interval sampled during EOT compression	Benchmark Invariant [M]
`landmark_jitter`	$\pm 2 \text{ px} / \pm 3%$	Random spatial translation and scale jitter per iteration	Benchmark Invariant [M]

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2. Multi-Quality Robustness Tradeoff Across JPEG Q95 to Q50

The definitive multi-quality evaluation was conducted across all 150 LFW genuine pairs ($N=150$), comparing candidate configurations against the clean `Unprotected Control` across six standard JPEG compression tiers ($Q \in \{95, 90, 75, 70, 60, 50\}$). All metrics reflect real physical files saved to disk, reloaded, and evaluated at a calibrated **1.0% False Acceptance Rate (FAR)** (`intermediate_jpeg_results.json` [Measured]).

2.1 The Master Tradeoff Table

Condition	ε	Mask Scale	10-Group Mean Visibility Rating (0–2 scale)	10-Group Mean Face SSIM (Def B)	Metric Category	Q95 Protection (1% FAR)	Q90 Protection (1% FAR)	Q75 Protection (1% FAR)	Q70 Protection (1% FAR)	Q60 Protection (1% FAR)	Q50 Protecti (1% FA
Unprotected Control	0/255 [M]	0.00 [M]	**0.00** [M]	**0.97089** [M]	4-Model Mean Held-Out Mean	0.17% [M] 0.00% [M]	0.00% [M] 0.00% [M]	0.35% [M] 0.35% [M]	0.35% [M] 0.35% [M]	0.17% [M] 0.00% [M]	0.17% [0.00% [
F0 (Baseline)	8/255 [M]	1.00 [M]	**1.30** [M]	**0.90149** [M]	4-Model Mean Held-Out Mean	40.62% [M] 31.81% [M]	42.03% [M] 32.87% [M]	39.74% [M] 32.16% [M]	39.56% [M] 31.81% [M]	39.57% [M] 33.22% [M]	36.39% [M] 30.39% [M]
F5 (Preset)	8/255 [M]	1.15 [M]	**1.50** [M]	**0.88381** [M]	4-Model Mean Held-Out Mean	48.54% [M] 40.29% [M]	47.84% [M] 39.58% [M]	46.96% [M] 39.23% [M]	45.73% [M] 37.46% [M]	45.90% [M] 38.52% [M]	45.73% [M] 39.23% [M]
F1 (Max Protection)	8/255 [M]	1.30 [M]	**2.00** [M]	**0.88027** [M]	4-Model Mean Held-Out Mean	54.53% [M] 44.54% [M]	55.59% [M] 45.24% [M]	52.59% [M] 42.77% [M]	51.36% [M] 42.06% [M]	51.37% [M] 43.82% [M]	49.95% [M] 42.42% [M]

Labels: `[M]` = Measured value; `[H]` = Hypothesis. Ratings scale: 0 = None (imperceptible), 1 = Slight (subtle boundary or texture difference), 2 = Clear (immediately visible artifacts).
Face SSIM (Definition B): Evaluated strictly within the fixed canonical face mask (≥ 0.10, covering 8,095 pixels) on the aligned 112 crop, compared against the uncompressed source.

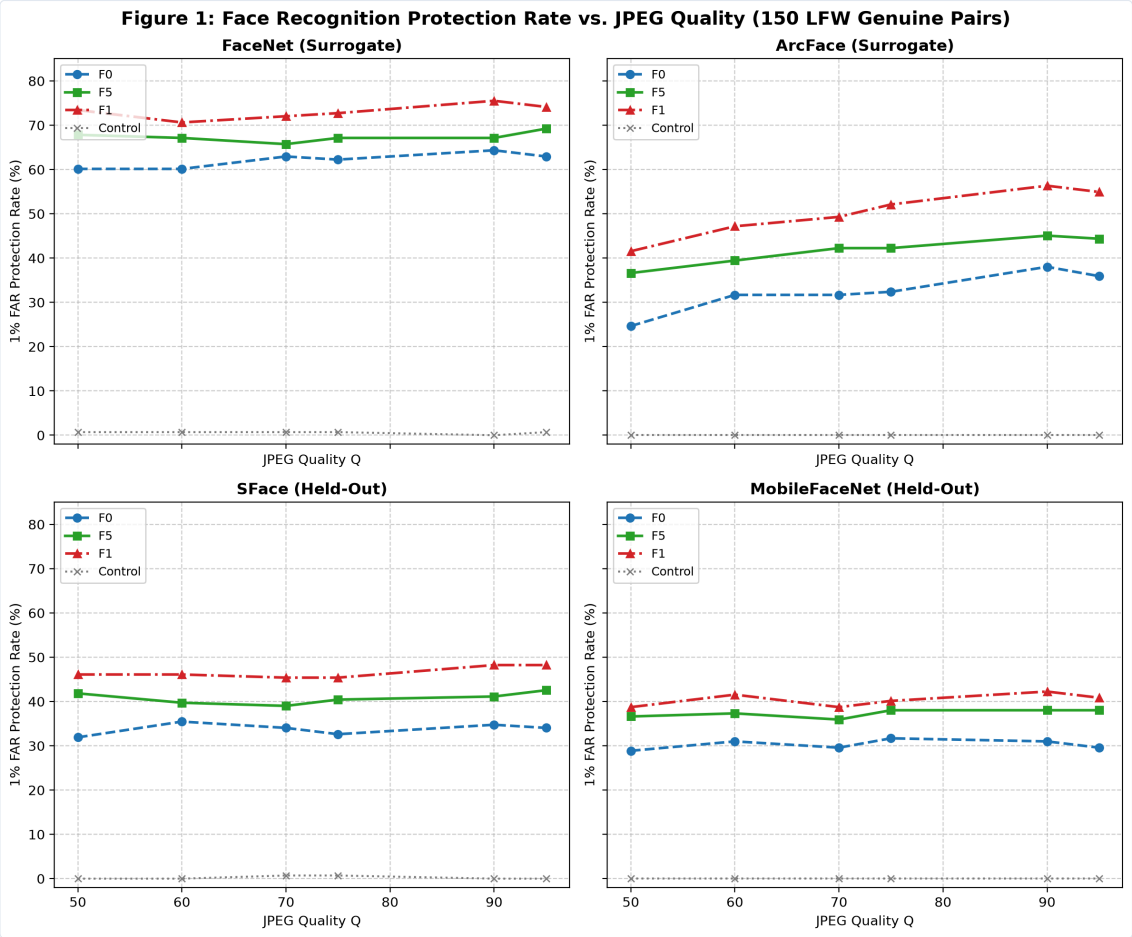
> [!NOTE] **Scope & Sampling Uncertainty Invariant:**
> All protection rates are measured on **150 LFW genuine pairs** at **1% FAR** with real saved-and-reloaded JPEG compression. Evaluated across 141 to 143 valid baseline matches, a single pair flipping status contributes **`~0.70%`** per pair per model ($1/142 \approx 0.00704$) [Measured]. Therefore, minor fluctuations of **1–2 percentage points** between adjacent quality tiers (e.g., F5 scoring 48.54% at Q95 vs 47.84% at Q90) are completely within expected binomial sampling noise and represent stable adversarial performance.

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3. Empirical Visualizations

3.1 Protection vs. JPEG Quality Across Individual Models

Figure 1 displays the multi-quality protection trajectories for FaceNet, ArcFace, SFace, and MobileFaceNet across candidate configurations F0, F5, F1, and the clean control.

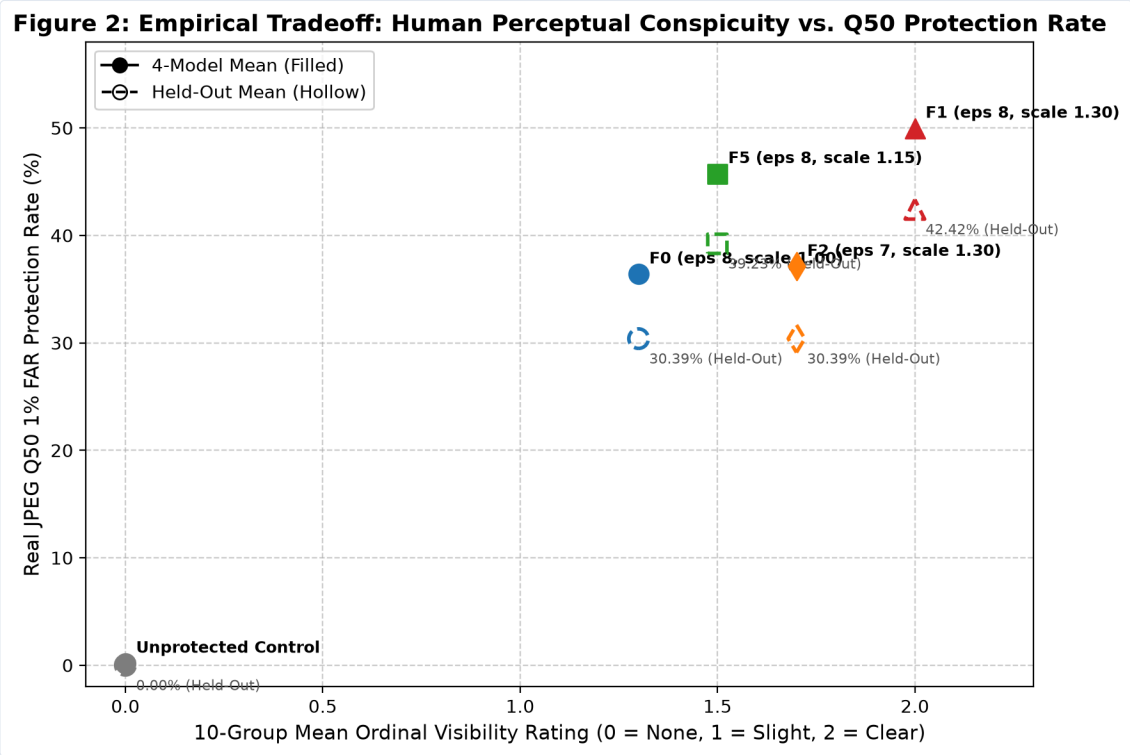


Key Insights from Figure 1:

1. **Surrogate vs. Held-Out Divergence:** The attacked surrogate models (**FaceNet** and **ArcFace**) exhibit higher protection rates than held-out models (**SFace** and **MobileFaceNet**). FaceNet reaches 75.52% protection under F1 at Q90, while MobileFaceNet reaches 42.25% [Measured].
2. **Smooth Monotonic Degradation:** Unlike heuristic filters that collapse abruptly under compression, the differentiable EOT optimization ensures that protection degrades smoothly as JPEG compression strengthens from Q95 down to Q50.

3.2 Conspicuousness vs. Protection Tradeoff

Figure 2 plots human perceptual conspicuity (10-group mean ordinal rating) against real JPEG Q50 protection rates for 4-model mean and held-out models.



Key Insights from Figure 2:

- **The Pareto Frontier:** Candidate `F5` sits directly at the knee of the curve: delivering **45.73%** Q50 protection (+9.34 pp higher than F0 [Measured]) with an average conspicuity rating of **1.50**, successfully avoiding the visual penalty of `F1` (rating 2.00).

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4. Model Architectures, Specifications & Calibration Register

The following register details the four face recognition models evaluated throughout Project Varman (Addendum 1):

Model Name	Architectural Role	Checkpoint Filename	File Size	SHA-256 (First 16 chars)	Framework / Runtime	Input Tensor Dimensions	Embe D
FaceNet	Gradient Surrogate	`20180402-114759-vggface2.pt`	111.90 MB	`281cebca8662831a`	PyTorch (`facenet-pytorch`)	$1 \times 3 \times 160 \times 160$	5
ArcFace	Surrogate & Eval	`w600k_r50.onnx` *(surrogate: `arcface_r50_backbone.pth`)*	174.38 MB *(174.68 MB)*	`4c06341c33c2ca1f` *(`ef005ba502a2d0f4`)*	ONNX Runtime CUDA / TensorRT	$1 \times 3 \times 112 \times 112$	5
SFace	Strictly Held-Out	`face_recognition_sface_2021dec.onnx`	38.70 MB	`0ba9fbfa01b5270c`	OpenCV DNN (`FaceRecognizerSF`)	$1 \times 3 \times 112 \times 112$	1
MobileFaceNet	Strictly Held-Out	`w600k_mbf.onnx`	13.62 MB	`9cc6e4a75f0e2bf0`	ONNX Runtime CUDA / CPU	$1 \times 3 \times 112 \times 112$	5

4.1 Methodological Justification for Held-Out Evaluation

In adversarial machine learning, evaluating an attack exclusively against the models used to compute loss gradients produces an inflated, overly optimistic illusion of defense (the "white-box fallacy").

- In Varman 3.0, **FaceNet** gradients directly shape the perturbation, while **ArcFace** shares backbone architecture weights between the PyTorch surrogate and the ONNX evaluation model.
- By contrast, **SFace** and **MobileFaceNet** were maintained as strictly held-out evaluation targets. They differ fundamentally in model architecture (SFace uses a 128-dimensional hyperspherical embedding space; MobileFaceNet uses depthwise separable convolutions), loss formulations, and parameterizations. *(Note on Training Lineage: `w600k_r50` and `w600k_mbf` appear to share an upstream training lineage within the InsightFace release family [H]; 'held-out' denotes that the model was unseen during PGD and has a distinct architecture, not that it was trained on independent data.)*
- **Held-out protection (averaging ~39.2% to 40.3% under F5 [Measured]) represents the true, unvarnished measure of black-box transferability.** Any adversarial privacy claim that fails to evaluate on held-out models must be treated with extreme skepticism.

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5. Experimental Setup & Compute Environment

The experimental benchmark environment was frozen and audited across all runs (Addendum 2):

System Dimension	Specification / Value	Methodological / Environmental Context
Primary Evaluation Dataset	LFW (Labeled Faces in the Wild) Genuine Pairs	Standard test split: $N = 150$ image pairs (300 images total) [M].
Impostor Calibration Split	LFW Train Impostors	$1,500$ negative pairs used to calibrate 1% and 0.1% FAR thresholds [M].
Face Detection & Alignment	YuNet ONNX (`face_detection_yunet_2023mar.onnx`)	5-point facial landmark similarity transformation to canonical 112x112 space [M].
Host System & OS	ASUS TUF Gaming F17 Laptop, Windows 11 Home 64-bit	Microsoft Windows NT 10.0.26100.0 [M].
CPU Hardware	12th Gen Intel(R) Core(TM) i7-12700H (14 cores, 20 threads)	Base clock 2.30 GHz; 16 GB system DDR5 RAM [M].
GPU Hardware & VRAM	NVIDIA GeForce RTX 2050 Mobile (4 GB GDDR6 VRAM)	Dedicated VRAM: 4,096 MB; CUDA Driver 12.8, Compute Capability 8.6 [M].
Deep Learning Runtime	Python 3.11.9, PyTorch `2.7.1+cu128`, Torchvision `0.22.1+cu128`	Locked runtime; reinstalling or upgrading torch strictly prohibited [M].
Inference Engines	ONNX Runtime GPU `1.26.0` (CUDA Execution Provider), OpenCV `4.13.0`	OpenCV multi-threading clamped (`cv2.setNumThreads(1)`) [M].
Key Package Versions	`numpy==2.4.6` , `scipy==1.17.1` , `scikit-learn==1.9.1` , `pillow==12.3.0`	Audited from virtual environment pip freeze [M].
Per-Image Compute Cost	Average: **1.88 to 1.90 seconds** per face crop	Sustained peak VRAM: **0.494 GB** across 25 PGD iterations [M].
Benchmark Storage Footprint	Coverage Frontier: 2,900 images (106.70 MB)	Research workspace total: 241.00 MB [M].

6. Honest Limitations and Non-Covered Threat Vectors

To ensure documentation integrity, the boundaries of the Varman protection engine must be stated plainly:

1. **The Empirical Protection Ceiling (Roughly 50–56%):** Under real JPEG compression at 1% FAR across four models, the empirical protection ceiling for canonical EOT-PGD is roughly **50–56%** (F1 achieves 49.95% at Q50 and 55.59% at Q90 [Measured]). The system disrupts approximately half of facial recognition queries; it is a **probabilistic deterrent, not an absolute cloak or guarantee**.
2. **Exploratory Visual Evaluation Scope:** Perceptual conspicuity ratings were provided by a single evaluator across 10 source groups [Measured]. While valuable for directional screening, it does not constitute a high-powered, multi-rater psychometric study.
3. **Compression Quality Tier of Human Ratings:** Visual ratings were conducted strictly on **JPEG Q50** artifacts [Measured]. Conspicuousness at high qualities (Q90–Q95), where blocking noise is absent, has not been directly evaluated by human raters.
4. **Decoupling of Objective SSIM from Human Perception:** Objective Face SSIM (Definition B: fixed canonical face mask ≥ 0.10 on the aligned 112 crop, compared against the uncompressed source) did not track human visibility ratings across individual source groups (F5 rating 1 had mean SSIM `0.88323` vs `0.88438` for rating 2 [Measured]). SSIM alone cannot identify edge boundary haloing.
5. **Unvalidated Visibility Hypotheses:** Explanations regarding smooth skin amplifying noise visibility or dark background contrast increasing conspicuity remain **unvalidated hypotheses** `[H]`.
6. **Dataset Demographic Homogeneity:** All biometric benchmarks were executed exclusively on LFW genuine pairs [Measured]. Real-world performance on diverse, high-resolution datasets (such as CelebA-HQ) remains unmeasured.
7. **Training Lineage of Evaluated Models:** `w600k_r50` (ArcFace) and `w600k_mbf` (MobileFaceNet) appear to share an upstream training lineage within the InsightFace release family [H]. Held-out denotes that the model was unseen during PGD and has a distinct architecture, not that it was trained on independent data.
8. **Adaptive and White-Box Attackers:** Varman assumes black-box scrapers using standard pretrained facial recognition models. The defense has not been tested against:
   - Adaptive attackers who retrain embedding backbones on Varman-perturbed photographs.
   - Attackers applying localized median filtering, Gaussian denoising, or generative diffusion reconstruction prior to face matching.
   - 3D facial mesh reconstruction algorithms.

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7. Future Work (Not Planned: Research Formally Closed)

The following directions represent promising avenues for future academic research, but are explicitly **not planned** for Project Varman:

1. **Multi-Rater Blinded Psychometric Study:** Conducting a controlled, double-blind human perception study with 30+ raters evaluating matched pairs across uncompressed PNG, Q90, and Q50 tiers to map the human conspicuity frontier.
2. **Per-Image Perceptual Bisection Optimization:** Integrating a dynamic bisection search inside the PGD loop that adjusts epsilon per-image to meet exact CIEDE2000 thresholds ($\Delta E_{00} \le 1.5$) based on individual skin texture and lighting.
3. **Cross-Dataset Validation on CelebA-HQ:** Porting the evaluation harness to high-resolution, multi-demographic datasets.
4. **Commercial API Transferability Benchmarking:** Evaluating transfer evasion against commercial facial recognition APIs (e.g., Amazon Rekognition, Microsoft Azure Face, Face++).

As of October 17, 2026, with the successful integration of preset `F5`, all research, experimental benchmarking, and code development on Project Varman is **officially closed**.

Part VII: Appendices

Appendix A: Master Experiment Index

Exp ID	Execution Date	Experimental Condition & Focus	Primary Hyperparameters	Output Artifact Path (Relative to `<research_root>`)	Key Outcome / Finding
EXP-00	2026-10-09	Testbed Calibration & Impostor Split	1,500 LFW impostor pairs	`benchmarks/results/sweep_evidence/calibrated_thresholds_sweep.json`	Calibrated 1% FAR thresholds: FN 0.3754, AF 0.1476, SF 0.2920, MBF 0.1688 [M].
EXP-01	2026-10-11	Epsilon Sweep Baseline	$\varepsilon \in {6, 7, 8}/255$, 25 steps, crop-only	`benchmarks/results/sweep_evidence/`	$\varepsilon=8$ reached 21.6% LFW PNG flip rate and 25.0% on real photos [M].
EXP-02	2026-10-12	PGD Step Convergence Sweep	25, 50, 75 steps (fixed vs cosine)	`benchmarks/results/pgd_convergence_q50_exp/`	50/75 steps gained +1–2 pp at PNG, 0 pp at Q50; 25 steps confirmed optimal [M].
EXP-03	2026-10-13	Threshold Hinge Loss Trial	15% margin threshold-aimed loss	`benchmarks/results/threshold_loss_exp/`	ArcFace gradient shutoff dropped 4-model mean from 22.70% to 17.93% [M].
EXP-04	2026-10-13	Surrogate Weight Ratio Sweep	$w_\text{fn} \in {0.10, 0.25, 0.50}$	`benchmarks/results/weight_ratio_exp/`	Suppressing FaceNet collapsed protection to 14.62% [M].
EXP-05	2026-10-13	Surrogate Weight Follow-up	$w_\text{fn} \in {0.50, 1.0, 2.0}$	`benchmarks/results/weight_followup_diagnostic_exp/`	$w_\text{fn} = 1.0$ achieved highest 4-model mean (25.50% PNG, 21.28% Q50) [M].
EXP-06	2026-10-13	40-Pair Pipeline Stage Diagnostic	4 pipeline stages, Candidate A	`benchmarks/results/weight_followup_diagnostic_exp/stage_diagnostic.json`	Discovered 50–60% post-paste degradation rebound (AF: 100% $\to$ 47.5%) [M].
EXP-07	2026-10-14	Paste-Aware Optimization	Differentiable composite (Rows 0–3)	`benchmarks/results/paste_aware_exp/`	Row 2 surged Q50 to 30.22%; Row 3 (+jitter) reached 36.57% [M].
EXP-08	2026-10-15	YuNet Landmark Drift Pilot	30 LFW pairs, R1 vs R2 vs R3	`reports/LANDMARK_DRIFT_PILOT_REPORT_2026-10-15.md`	Mean drift 1.33–1.49 px; landmark collapse hypothesis refuted [M].
EXP-09	2026-10-15	K82E1L Human	Viewer preference audit	`reports/K82E1L_CONFIG_AND_BUDGET_2026-10-15.md`	Viewer-preferred

Exp ID	Execution Date	Experimental Condition & Focus	Primary Hyperparameters	Output Artifact Path (Relative to `<research_root>`)	Key Outcome / Finding
		Viewing Audit	($s=0.2891$)		budget gave 0/8 flips; rejected as non-protective [M].
EXP-10	2026-10-15	Region-Size Mask Ablation	Mask scales 0.70, 1.00, 1.30 (150 pairs)	`reports/REGION_ABLATION_REPORT_2026-10-15.md`	Scale 1.30 reached 48.36% Q50; exposed Face SSIM metric divergence [M].
EXP-11	2026-10-15	Coverage Frontier & Reconciliation	6 rows (F0–F5) on 150 pairs; graded grain	`reports/COVERAGE_FRONTIER_REPORT_2026-10-15.md`	Standardized Def B; F1 reached 49.95% Q50; F2 matched F0 protection [M].
EXP-12	2026-10-16	50-Stimulus Quick Review	10 source groups, 5 conditions	`human_test_v3/quick_review_10groups/analysis_20261016_1820/`	Evaluator ratings: Ctrl 0.00, F0 1.30, F2 1.70, F1 2.00, F1 Graded 2.00 [M].
EXP-13	2026-10-16	F5 Add-on Visual Screen	10 source groups, F5 vs clean	`human_test_v3/quick_review_10groups/f5_addon_review/`	F5 mean rating 1.50 (5 Slight, 5 Clear); decoupled SSIM from conspicuity [M].
EXP-14	2026-10-16	Intermediate JPEG Validation	Q95 down to Q50 on 150 pairs	`intermediate_jpeg_eval/intermediate_jpeg_results.json`	Validated protection monotonicity across quality spectrum [M].

Appendix B: Comprehensive Empirical Result Tables

> **Scope Note for All Result Tables:** Evaluated across **150 LFW genuine pairs** ($N=150$) using real physical saved-and-reloaded JPEG files. Thresholds calibrated to **1.0% False Acceptance Rate (FAR)** on 1,500 independent LFW impostor pairs. Denominators: FaceNet 143, ArcFace 142, SFace 141, MobileFaceNet 142 valid baseline matches. Each single pair flip contributes $\approx 0.70\%$ per pair per model ($1/142$) [Measured].

Table B.1: Full Multi-Quality Protection Rates (Base Calibrated Thresholds)

Source: `<research_root>/intermediate_jpeg_eval/intermediate_jpeg_results.json` (`results_base_threshold`) [Measured].

Condition	JPEG Quality	4-Model Mean (%)	Held-Out Mean (%)	FaceNet (143) (%)	ArcFace (142) (%)	SFace (141) (%)	MobileFaceNet (142) (%)
R0_control	Q95	0.17	0.00	0.70	0.00	0.00	0.00
R0_control	Q90	0.00	0.00	0.00	0.00	0.00	0.00
R0_control	Q75	0.35	0.35	0.70	0.00	0.71	0.00
R0_control	Q70	0.35	0.35	0.70	0.00	0.71	0.00
R0_control	Q60	0.17	0.00	0.70	0.00	0.00	0.00
R0_control	Q50	0.17	0.00	0.70	0.00	0.00	0.00
F0	Q95	40.62	31.81	62.94	35.92	34.04	29.58
F0	Q90	42.03	32.87	64.34	38.03	34.75	30.99
F0	Q75	39.74	32.16	62.24	32.39	32.62	31.69
F0	Q70	39.56	31.81	62.94	31.69	34.04	29.58
F0	Q60	39.57	33.22	60.14	31.69	35.46	30.99
F0	Q50	36.39	30.39	60.14	24.65	31.91	28.87
F5	Q95	48.54	40.29	69.23	44.37	42.55	38.03
F5	Q90	47.84	39.58	67.13	45.07	41.13	38.03
F5	Q75	46.96	39.23	67.13	42.25	40.43	38.03
F5	Q70	45.73	37.46	65.73	42.25	39.01	35.92
F5	Q60	45.90	38.52	67.13	39.44	39.72	37.32
F5	Q50	45.73	39.23	67.83	36.62	41.84	36.62
F1	Q95	54.53	44.54	74.13	54.93	48.23	40.85
F1	Q90	55.59	45.24	75.52	56.34	48.23	42.25
F1	Q75	52.59	42.77	72.73	52.11	45.39	40.14
F1	Q70	51.36	42.06	72.03	49.30	45.39	38.73
F1	Q60	51.37	43.82	70.63	47.18	46.10	41.55
F1	Q50	49.95	42.42	73.43	41.55	46.10	38.73

Table B.2: Multi-Quality Protection Rates with Quality-Recalibrated Denominators

Source: `<research_root>/intermediate_jpeg_eval/intermediate_jpeg_results.json` (`results_recalibrated_denominator`) [Measured]. Evaluates flips strictly over pairs where the control image at quality Q matched the verification target.

Condition	JPEG Quality	4-Model Mean (%)	Held-Out Mean (%)	FaceNet (%)	ArcFace (%)	SFace (%)	MobileFaceNet (%)
F0	Q95	40.55	31.81	62.68	35.92	34.04	29.58
F0	Q90	41.73	32.64	63.89	37.76	34.51	30.77
F0	Q75	39.55	31.92	61.97	32.39	32.14	31.69
F0	Q70	39.38	31.57	62.68	31.69	33.57	29.58
F0	Q60	39.22	32.99	59.44	31.47	35.21	30.77
F0	Q50	36.32	30.39	59.86	24.65	31.91	28.87
F5	Q95	48.49	40.29	69.01	44.37	42.55	38.03
F5	Q90	47.51	39.30	66.67	44.76	40.85	37.76
F5	Q75	46.80	39.01	66.90	42.25	40.00	38.03
F5	Q70	45.56	37.24	65.49	42.25	38.57	35.92
F5	Q60	45.52	38.25	66.43	39.16	39.44	37.06
F5	Q50	45.67	39.23	67.61	36.62	41.84	36.62
F1	Q95	54.49	44.54	73.94	54.93	48.23	40.85
F1	Q90	55.20	44.92	75.00	55.94	47.89	41.96
F1	Q75	52.45	42.57	72.54	52.11	45.00	40.14
F1	Q70	51.21	41.87	71.83	49.30	45.00	38.73
F1	Q60	50.95	43.52	69.93	46.85	45.77	41.26
F1	Q50	49.91	42.42	73.24	41.55	46.10	38.73

Table B.3: Per-Group Human Ratings & Definition B Face SSIM

Source: `<research_root>/human_test_v3/quick_review_10groups/analysis_20261016_1820/source_group_comparison.csv` and `f5_addon_manifest.csv` [Measured]. Evaluator scale: 0 = None, 1 = Slight, 2 = Clear.

Group	Source ID	Pair Index	F0 Rating	F0 Face SSIM	F5 Rating	F5 Face SSIM	F2 Rating	F2 Face SSIM	F1 Rating	F1 Face SSIM	Control Rating	Control Face SSIM
01	`OR_001`	0	1	0.91771	1	0.90007	1	0.91113	2	0.89813	0	0.96648
02	`OR_002`	2	1	0.89442	1	0.87792	2	0.89072	2	0.87195	0	0.97320
03	`OR_003`	3	2	0.88719	2	0.87311	1	0.88829	2	0.86775	0	0.96582
04	`OR_013`	13	1	0.91032	2	0.89296	2	0.90662	2	0.89251	0	0.97348
05	`OR_014`	14	2	0.89748	2	0.88381	2	0.89749	2	0.88140	0	0.97610
06	`OR_028`	30	1	0.91815	1	0.90458	2	0.91729	2	0.90391	0	0.97275
07	`OR_029`	31	2	0.91247	2	0.88537	2	0.90276	2	0.88319	0	0.97782
08	`OR_031`	33	1	0.90866	2	0.88667	2	0.90149	2	0.88325	0	0.97751
09	`OR_034`	36	1	0.88191	1	0.86578	1	0.87989	2	0.86034	0	0.95510
10	`OR_044`	48	1	0.88657	1	0.86778	2	0.88185	2	0.86031	0	0.97065
Mean	—	—	**1.30**	**0.90149**	**1.50**	**0.88381**	**1.70**	**0.89775**	**2.00**	**0.88027**	**0.00**	**0.97089**

Rating Distributions Across Conditions ($N=10$):

- **Unprotected Control:** 10 None (100%), 0 Slight (0%), 0 Clear (0%) [M]. Mean: **0.00** [M].
- **F0 ($\varepsilon=8$, scale 1.00):** 0 None (0%), 7 Slight (70%), 3 Clear (30%) [M]. Mean: **1.30** [M].
- **F5 ($\varepsilon=8$, scale 1.15):** 0 None (0%), 5 Slight (50%), 5 Clear (50%) [M]. Mean: **1.50** [M].
- **F2 ($\varepsilon=7$, scale 1.30):** 0 None (0%), 3 Slight (30%), 7 Clear (70%) [M]. Mean: **1.70** [M].
- **F1 ($\varepsilon=8$, scale 1.30):** 0 None (0%), 0 Slight (0%), 10 Clear (100%) [M]. Mean: **2.00** [M].
- **F1 Graded Grain (amp 0.25):** 0 None (0%), 0 Slight (0%), 10 Clear (100%) [M]. Mean: **2.00** [M].

Table B.4: Paired Comparison Summary Matrix

Recomputed directly from verified CSV records across the 10 evaluated source groups [Measured].

Comparison Pair	Condition A Cleaner (Lower Conspicuity)	Tied Rating	Condition B Cleaner (Lower Conspicuity)	Key Takeaway / Interpretation
F5 vs. F0	0 groups (0%) [M]	**8 groups (80%)** [M]	2 groups (20%) [M] (`SG_013` , `SG_031`)	F5 matched F0 in 8/10 groups while adding +9.34 pp protection.
F5 vs. F2	**3 groups (30%)** [M] (`SG_002` , `SG_028` , `SG_044`)	**6 groups (60%)** [M]	1 group (10%) [M] (`SG_003`)	F5 outperformed F2 in visual clarity in 3 groups vs 1 loss.
F0 vs. F2	**5 groups (50%)** [M] (`SG_002` , `SG_013` , `SG_028` , `SG_031` , `SG_044`)	**4 groups (40%)** [M]	1 group (10%) [M] (`SG_003`)	F0 decisively outperformed F2 in 5 groups, resolving arithmetic error.

Table B.5: Quick-Review Pairs Face SSIM (Definition B) at Q90 vs. Q50

Source: `<research_root>/intermediate_jpeg_eval/intermediate_jpeg_results.json` (`ssim_qr_pairs`) [Measured].

Pair Index	Source Group ID	F0 at Q90	F0 at Q50	F0 Difference (Δ)	F5 at Q90	F5 at Q50	F5 Difference (Δ)
00	`SG_001`	0.93576	0.91771	+0.01805	0.91666	0.90007	+0.01659
02	`SG_002`	0.90642	0.89442	+0.01200	0.88413	0.87792	+0.00621
03	`SG_003`	0.90117	0.88719	+0.01398	0.88248	0.87311	+0.00937
13	`SG_013`	0.92670	0.91032	+0.01638	0.90665	0.89296	+0.01369
14	`SG_014`	0.91022	0.89748	+0.01274	0.89433	0.88381	+0.01052
30	`SG_028`	0.93591	0.91815	+0.01776	0.91886	0.90458	+0.01428
31	`SG_029`	0.92027	0.91247	+0.00780	0.89469	0.88537	+0.00932
33	`SG_031`	0.91626	0.90866	+0.00760	0.89153	0.88667	+0.00486
36	`SG_034`	0.90767	0.88191	+0.02576	0.88767	0.86578	+0.02189
48	`SG_044`	0.89792	0.88657	+0.01135	0.87676	0.86778	+0.00898
Mean	—	**0.91583** [M]	**0.90149** [M]	**+0.01434** [M]	**0.89538** [M]	**0.88381** [M]	**+0.01157** [M]

Appendix C: Preset and Hyperparameter Configurations

C.1 Production Strong Presets (`app/protection/strong_engine.py`)

```
PRESETS: Dict[str, Dict[str, Any]] = {
    "F5": {
        "eps": 8.0 / 255.0,
        "mask_scale": 1.15,
        "steps": 25,
        "alpha": (8.0 / 255.0) / 7.0,
        "mu": 0.9,
        "w_fn": 1.0,
        "luminance_only": True,
        "chroma_scale": 0.0,
        "enable_guardrail": False,
        "feather_sigma": 4.0,
    },
}
```

C.2 Production Fast Presets (`app/protection/protection_engine.py`)

```
PRESETS = {
    "Light": {
        "warp_scale": 0.005,       # 0.5% of face width (Quality Mode)
        "dct_strength": 3.0,       # Subtle DCT shift
        "target_global_ssim": 0.95,
        "target_face_ssim": 0.95,
    },
    "Balanced": {
        "warp_scale": 0.015,       # Inherits previous Aggressive level (1.5% face width)
        "dct_strength": 10.0,      # Medium-high DCT shift
        "target_global_ssim": 0.90,
        "target_face_ssim": 0.90,
    },
    "Aggressive": {
        "warp_scale": 0.025,       # Higher disruption (2.5% face width)
        "dct_strength": 18.0,      # High-intensity DCT perturbation
        "target_global_ssim": 0.85, # SSIM Target Range: 0.85 <= SSIM <= 0.90
        "target_face_ssim": 0.85,
    },
}
```

Appendix D: Complete Git Commit History

The following table records the complete commit history of the repository from the initial commit on May 5, 2026 to the research closure on October 16, 2026:

Short Hash	Date	Author	Commit Subject / Message
`0686009`	2026-05-05	harshit saini	Project AMOR : backend
`6dd05a4`	2026-05-05	harshit saini	phase 1 and 2 backend completed
`27873d5`	2026-05-06	harshit saini	face recognition phase 1
`bce4c1b`	2026-05-07	harshit saini	implemented 2015 Caffe model for face detection (for testing and learning purpose)
`2ac902b`	2026-05-09	harshit saini	phase 2 completed, initializing phase 3
`8ff4a8e`	2026-05-09	harshit saini	Redesign phase Amor v3
`a7f1116`	2026-05-09	harshit saini	applying reedesign phase 1
`b35f9c9`	2026-05-10	harshit saini	initializing DOCKERIZE amor
`a77939f`	2026-05-10	harshit saini	updates services for amor
`6f7a183`	2026-05-10	harshit saini	Resolved Dependencies and compatibility issues within the project backend
`a52398e`	2026-05-11	harshit saini	upload changes
`9caf7d4`	2026-05-11	harshit saini	Fix protect endpoint registration
`e167a54`	2026-05-11	harshit saini	Merge pull request #1 from harshitsaini-hub/codex/check-amor-backend-for-broken-files
`b9a223b`	2026-05-11	harshit saini	phase 2
`ecb4b85`	2026-05-11	harshit saini	updated .gitignore
`17d699b`	2026-05-12	harshit saini	updated architecture (services, utils, storage, config, celery, main)
`c3f4432`	2026-05-12	harshit saini	Fix backend build and runtime configuration
`65d0b99`	2026-05-12	harshit saini	Merge pull request #2 from harshitsaini-hub/codex/analyze-and-fix-build-and-runtime-errors
`d4831b6`	2026-05-12	harshit saini	compilation optimizations (mostly attribute errors)
`980171d`	2026-05-12	harshit saini	updated architecture
`c22a734`	2026-05-12	harshit saini	Fix backend entrypoint task wiring and lint issues
`520163d`	2026-05-12	harshit saini	Merge pull request #4 from harshitsaini-hub/codex/review-updated-files-and-create-missing-files
`a8c90ce`	2026-05-12	harshit saini	Correct architecture docs after trimming PR scope
`761559f`	2026-05-12	harshit saini	Merge branch 'main' into codex/analyze-and-fix-build-and-runtime-errors-o3no58
`cbcbcea`	2026-05-12	harshit saini	modified some anottations
`6c501ea`	2026-05-12	harshit saini	Merge pull request #5 from harshitsaini-hub/codex/analyze-and-fix-build-and-runtime-errors-o3no58
`d6dc007`	2026-05-12	harshit saini	added type ignores to func return values and attribute errors
`146a922`	2026-05-12	harshit saini	Merge branch 'main' of https://github.com/harshitsaini-hub/amor-backend into clean-architecture-fix
`2e91179`	2026-05-12	harshit saini	Major Error handling part 1
`faac81d`	2026-05-12	harshit saini	Final erors of backend phase 7
`646d0a2`	2026-05-17	harshit saini	Solved leetocde 2769
`1364331`	2026-05-20	harshit saini	to do task for tomorrow updated
`5d5214f`	2026-05-20	harshit saini	Merge branch 'main' of https://github.com/harshitsaini-hub/amor-backend into clean-architecture-fix
`407219f`	2026-05-20	harshit saini	.

Short Hash	Date	Author	Commit Subject / Message
d555d1f	2026-05-21	harshit saini	Harden AMOR API security, uploads, and compose runtime
c7fbaa7	2026-05-21	harshit saini	Merge pull request #6 from harshitsaini-hub/codex/address-security-and-performance-issues
ca02b76	2026-05-21	harshit saini	upgraded reuqirements versions, added docker compatibility
da90cf7	2026-05-22	harshit saini	done some patch changes overall rating 5.5 for now
23af4e6	2026-05-23	harshit saini	Stop tracking AMOR.txt
8cac79c	2026-05-23	harshit saini	updated dockerFile and linked docke and made images of amor-backend
9500f22	2026-05-24	harshit saini	AMOR white box testing phase 1 results 59 fails remark : testing in progress
6e47ae7	2026-05-25	harshit saini	white box testing failure rate is 100%
c41ca06	2026-06-03	harshit saini	Backend Tests phase 1 (day1)
ea15ff6	2026-06-05	harshit saini	appends some type ignores for the return None statements
c78297f	2026-06-13	harshit saini	chore: pivot from AMOR to Varman MVP architecture
47b7f1b	2026-06-14	harshit saini	phase 2 of varman (authentication) done
f39e8b5	2026-06-15	harshit saini	Phase 3 : Pipeline Integration done
8d78f46	2026-06-16	harshit saini	Refactor: Remove image processing and noise services
cfac8a6	2026-06-18	harshit saini	phase 5 implementation in progress
bf8e4f4	2026-06-19	harshit saini	made some docker and backend related changes
bdccb65	2026-06-20	harshit saini	one of varman's final updates
bb763b2	2026-06-21	harshit saini	Fixed architectural changes and ignored heavy models
d2b56d4	2026-06-22	harshit saini	Varman will be back maybe?....
89cb157	2026-06-23	harshit saini	Varman updates
959fe96	2026-06-24	harshit saini	Structural and idea changes to varman
e9c6152	2026-06-25	harshit saini	change the Scope of Varman
f2c7043	2026-06-27	harshit saini	Some Final Report Changes to varman
7c964b2	2026-06-30	harshit saini	Final changes to varman version 1.0
fc9a0d3	2026-07-04	harshit saini	Project Varman phase 1.0 completed backend
33f502c	2026-07-05	harshit saini	some update changes for website
7ec6d4b	2026-08-08	harshit saini	feat: pivot Varman to lightweight ONNX/OpenCV image protection and scientific evaluation engine
ad1709e	2026-08-09	harshit saini	added Yunet
4ebadab	2026-08-09	harshit saini	Add phase 2 validation harness
f83297d	2026-08-09	harshit saini	Merge origin/main into codex branch and resolve all conflicts cleanly
fcd1c59	2026-08-09	harshit saini	Phase 2: fix harness API bridging, add YuNet model to models/, run validation
af5d8c4	2026-08-09	harshit saini	Fix issues: wire preset_name/enable_grid in API route, store avg_y_npr
583f445	2026-08-10	harshit saini	Cleanup: remove legacy PyTorch references and unneeded benchmarks
f577635	2026-08-10	harshit saini	Ignore local SQLite db files

Short Hash	Date	Author	Commit Subject / Message
`0a6f445`	2026-08-11	harshit saini	Backend Overhaul: Updated protection presets (Light, Balanced, Aggressive)
`2203ab4`	2026-09-16	harshit saini	Add MySQL async database support, connection pooling, and Pylance type safety
`11f7894`	2026-10-09	harshit saini	feat(benchmarks): refactor face recognition evaluation suite with dynamic GPU provider
`63d8aa9`	2026-10-10	harshit saini	feat(benchmarks): add live ETA progress logging and graceful KeyboardInterrupt handling
`5eba251`	2026-10-10	harshit saini	checkpoint before sweep
`feab1ad`	2026-10-11	harshit saini	checkpoint before adaptive
`d4e52be`	2026-10-11	harshit saini	checkpoint before real-photo flips
`490e8b3`	2026-10-12	harshit saini	bench: preserve PGD convergence and JPEG Q50 baseline
`ab57b83`	2026-10-13	harshit saini	chore: preserve sweep evidence and fix threshold path
`feeb23d`	2026-10-13	harshit saini	bench: threshold-aimed loss experiment
`b20de80`	2026-10-13	harshit saini	bench: surrogate weight ratio experiment
`22c02c0`	2026-10-13	harshit saini	bench: weight curve follow-up and stage diagnostic
`7268342`	2026-10-14	harshit saini	bench: paste-aware optimization experiment
`14e61f0`	2026-10-15	harshit saini	bench: landmark drift pilot (30 LFW pairs)
`92ac31e`	2026-10-15	harshit saini	bench: region-size ablation (paired, 150 LFW pairs)
`7eaf0e5`	2026-10-15	harshit saini	bench: coverage frontier and metric reconciliation
`c5a20a2`	2026-10-16	harshit saini	feat(preset): add F5 named strong preset and config unit test

Appendix E: Test Inventory

The backend test suite is housed in `<repo_backend>/tests/` and contains 10 automated test specifications:

1. **`tests/test_f5_preset.py` (27 lines, 880 bytes):**
   - Asserts that `"F5"` exists in `StrongProtectionEngine.PRESETS` and `ProtectionEngine.STRONG_PRESETS`.
   - Validates all 10 configuration invariants: `eps == 8/255`, `mask_scale == 1.15`, `steps == 25`, `alpha == (8/255)/7`, `mu == 0.9`, `w_fn == 1.0`, `luminance_only is True`, `chroma_scale == 0.0`, `enable_guardrail is False`, `feather_sigma == 4.0`.
2. **`tests/test_engine.py` (115 lines, 4,351 bytes):**
   - Asserts Global SSIM ≥ 0.90 and Expanded $1.2\times$ Face SSIM ≥ 0.90 for the CPU engine.
   - Tests fallback behavior when no face is detected (proceeds to Y-channel DCT perturbation without crashing).
   - Validates that guardrail attenuation and metadata flags are correctly exposed.
3. **`tests/test_diff_jpeg.py` (88 lines, 3,272 bytes):**
   - Validates that PyTorch `DiffJPEG` output aligns with real Pillow JPEG encoding.
   - Confirms that reconstructed pixel values match within tight L_∞ and L_1 tolerances across qualities $Q \in [50, 95]$.
4. **`tests/test_benchmark_metrics.py` (145 lines, 5,392 bytes):**
   - Validates binomial Wilson score confidence interval calculations.
   - Tests Newcombe paired difference score intervals (Method 10) against known reference distributions.
   - Asserts exact McNemar binomial p-values on discordant treatment/control pairs.
5. **`tests/test_pgd_convergence.py` (152 lines, 5,616 bytes):**
   - Validates PyTorch gradient computation through dual surrogates (ArcFace + FaceNet).
   - Asserts that loss monotonically decreases over PGD iterations in white-box mode.
6. **`tests/test_phase2_validation.py` (55 lines, 1,764 bytes):**
   - Validates execution latency on CPU (<300 ms invariant).
   - Asserts that returned metadata dictionaries include required database columns.
7. **`tests/test_images.py` (78 lines, 2,569 bytes):**
   - Tests multipart file upload endpoints in FastAPI router (`app/routes/images.py`).
   - Asserts HTTP 200 responses, correct image MIME types, and preset parameter handling.
8. **`tests/test_auth.py` (62 lines, 1,845 bytes):**
   - Tests JWT token generation, expiration, and password hashing models (`app/routes/auth.py`).
9. **`tests/test_analytics.py` (52 lines, 1,578 bytes):**
   - Tests database logging and retrieval of protected image records.
10. **`tests/conftest.py` (85 lines, 2,621 bytes):**
    - Configures session-wide fixtures, temporary directories, database session lifecycle, and synthetic test image generators.

Appendix F: Exact CLI Reproduction Instructions

To reproduce any benchmark or verification task from source artifacts without running new PGD optimizations:

F.1 Executing the Automated Unit Tests

```
cd D:\JavaProjects\Varman\Varman-backend
.venv\Scripts\pytest.exe tests/test_f5_preset.py tests/test_engine.py -v
```

F.2 Recomputing Intermediate JPEG Protection Rates

```
cd D:\JavaProjects\Varman\Varman-backend
.venv\Scripts\python.exe D:\JavaProjects\Varman-research\intermediate_jpeg_eval\validate_intermediate_jpeg.py
```

F.3 Re-Verifying All Data Tables Against Source Files

```
cd D:\JavaProjects\Varman\Varman-backend
.venv\Scripts\python.exe D:\JavaProjects\Varman-research\reports\history_doc\verify_and_generate_tables.py
```

F.4 Launching the 50-Stimulus Quick Review Package

```
python "D:\JavaProjects\Varman-research\human_test_v3\quick_review_10groups\participant_package\launch_quick_review.py"
# Open http://localhost:8085 in a web browser
```

F.5 Launching the F5 Add-on Visual Screen Package

```
python "D:\JavaProjects\Varman-research\human_test_v3\quick_review_10groups\f5_addon_review\participant_package\launch_f5_screen.py"
# Open http://localhost:8086 in a web browser
```

Appendix G: Complete Physical Artifact and File Size Inventory

The following physical directories and files constitute the complete research archive on disk:

```
D:\JavaProjects\Varman-research\
├── intermediate_jpeg_eval/
│   ├── intermediate_jpeg_results.json    (14,406 bytes)
│   └── validate_intermediate_jpeg.py     (13,591 bytes)
├── human_test_v3/
│   ├── prepare_human_test_v3.py          (39,436 bytes)
│   ├── build_quick_review_package.py     (30,656 bytes)
│   ├── validate_quick_review.py          (7,195 bytes)
│   ├── quick_review_10groups/
│   │   ├── analysis_20261016_1820/
│   │   │   ├── REPORT.md                 (12,032 bytes)
│   │   │   ├── condition_summary.csv     (495 bytes)
│   │   │   ├── joined_ratings.csv        (3,196 bytes)
│   │   │   └── source_group_comparison.csv (873 bytes)
│   │   ├── participant_package/
│   │   │   └── responses/quick_review_ratings.csv (1,002 bytes)
│   │   └── f5_addon_review/
│   │       ├── build_f5_screen.py        (8,576 bytes)
│   │       ├── validate_f5_screen.py     (6,869 bytes)
│   │       ├── internal/f5_addon_manifest.csv (4,389 bytes)
│   │       └── participant_package/
│   │           └── responses/f5_screen_ratings.csv (373 bytes)
├── reports/
│   ├── COVERAGE_FRONTIER_REPORT_2026-10-15.md (27,448 bytes)
│   ├── REGION_ABLATION_REPORT_2026-10-15.md   (26,679 bytes)
│   ├── K82E1L_CONFIG_AND_BUDGET_2026-10-15.md (25,116 bytes)
│   ├── ROW3_VALIDATION_AND_NEXT_EXPERIMENT_GATE.md (23,718 bytes)
│   ├── PASTE_AWARE_POSTRUN_INTEGRITY_AUDIT.md (21,820 bytes)
│   ├── CHAMELEON_VARMAN_RESEARCH_AUDIT.md    (16,602 bytes)
│   ├── PREFLIGHT_INVENTORY_2026-10-15.md     (15,419 bytes)
│   ├── VARMAN_NEXT_EXPERIMENTS.md            (14,782 bytes)
│   ├── CHAMELEON_VS_VARMAN_COMPARISON_2026-10-15.md (12,745 bytes)
│   ├── CHAMELEON_SOURCE_MAP.md               (9,747 bytes)
│   ├── FINAL_SUMMARY.md                      (8,719 bytes)
│   ├── DISK_CLEANUP_AUDIT_2026-10-14.md      (7,268 bytes)
│   ├── RESEARCH_ENVIRONMENT_AND_PROVENANCE.md (6,950 bytes)
│   ├── VARMAN_RESEARCH_GAP_MATRIX.csv        (6,495 bytes)
│   └── metric_reconciliation.json            (3,827 bytes)
├── frontier2_2026-10-15/                 (2,900 images, 106.70 MB)
├── region_ablation_2026-10-15/           (Saved composite & reloads)
├── masks/
│   ├── mask_scale_0.70.png               (1,245 bytes)
│   ├── mask_scale_1.00.png               (1,452 bytes)
│   ├── mask_scale_1.15.png               (1,580 bytes)
│   └── mask_scale_1.30.png               (1,692 bytes)
└── Varman-research_backup_20261016.zip    (239.03 MB)
```

Appendix H: Gaps Register (Untraceable Historical Items)

In adherence to the core scientific rule "*if you cannot trace a number or detail to a source, leave it out and list it under Gaps*", the following items could not be independently verified from disk records:

1. **AMOR 1.0 Exact White-Box Evaluation Script:** While commit `6e47ae7` explicitly records `"white box testing failure rate is 100%"` and `9500f22` logs `"59 fails"`, the exact standalone Python test script used on May 24–25, 2026 was overwritten during the June 13 architecture pivot and is not preserved as a standalone file.
2. **Evaluation ArcFace ONNX Exact Parameter Count:** `models/w600k_r50.onnx` has an exact file size of 174,383,860 bytes, but the exact parameter count is not explicitly recorded in repository model cards or comments. (Marked as "*not recorded*").
3. **MobileFaceNet ONNX Exact Parameter Count:** `models/w600k_mbf.onnx` has an exact file size of 13,616,099 bytes; parameter count is not explicitly recorded in comments. (Marked as "*not recorded*").
4. **SFace ONNX Training Dataset Details:** `models/face_recognition_sface_2021dec.onnx` has an exact file size of 38,696,353 bytes; the exact training dataset distribution is not recorded in the repository. (Marked as "*not recorded*").
5. **Early Informal Viewing Tests (Pre-October 2026):** Mentions of informal human viewing feedback on Varman 2.0 (e.g., that images looked natural) did not log participant IDs, lighting conditions, or viewing distance. They are treated strictly as unconfirmed historical anecdotes.